

## 5-4

1. Let  $f$  be a function that is even and continuous on the closed interval  $[-3, 3]$ . The function  $f$  and its derivatives have the properties indicated in the table below.

| $x$      | 0         | $0 < x < 1$ | 1 | $1 < x < 2$ | 2         | $2 < x < 3$ |
|----------|-----------|-------------|---|-------------|-----------|-------------|
| $f(x)$   | 1         | Positive    | 0 | Negative    | -1        | Negative    |
| $f'(x)$  | Undefined | Negative    | 0 | Negative    | Undefined | Positive    |
| $f''(x)$ | Undefined | Positive    | 0 | Negative    | Undefined | Negative    |

Find the  $x$ -coordinate of each point at which  $f$  attains an absolute maximum value or an absolute minimum value. For each  $x$ -coordinate you give, state whether  $f$  attains an absolute maximum or an absolute minimum.

**Part A**

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at  $x = 0$

Absolute minimum at  $x = \pm 2$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns three of the following points:

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

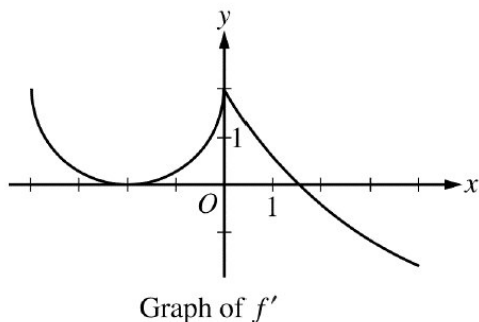
1 point is **deducted** for each additional point or incorrect point

Absolute maximum at  $x = 0$

Absolute minimum at  $x = \pm 2$

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2.



The derivative of a function  $f$  is defined by  $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$ .

The graph of the continuous function  $f$ , shown in the figure above, has  $x$ -intercepts at  $x = -2$  and  $x = 3 \ln\left(\frac{5}{3}\right)$ . The graph of  $g$  on  $-4 \leq x \leq 0$  is a semicircle, and  $f(0) = 5$ .

For  $-4 \leq x \leq 4$ , find the value of  $x$  at which  $f$  has an absolute maximum. Justify your answer.

**Part C**

1 point is earned for the answer

1 point is earned for the justification

Since  $f'(x) > 0$  on the intervals  $-4 < x < -2$  and  $-2 < x < 3 \ln\left(\frac{5}{3}\right)$ ,  $f$  is increasing on the interval  $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$ .

Since  $f'(x) < 0$  on the interval  $3 \ln\left(\frac{5}{3}\right) < x < 4$ ,  $f$  is decreasing on the interval  $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$ .

Therefore,  $f$  has an absolute maximum at  $x = 3 \ln\left(\frac{5}{3}\right)$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
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The student response earns two of the following points:

1 point is earned for the answer

1 point is earned for the justification

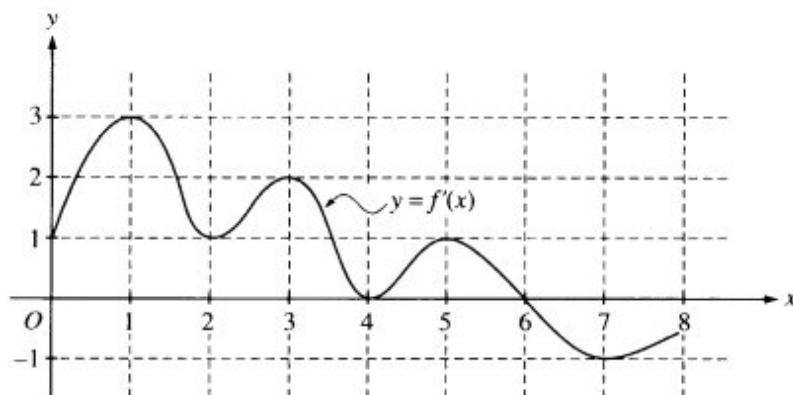
Since  $f'(x) > 0$  on the intervals  $-4 < x < -2$  and  $-2 < x < 3 \ln\left(\frac{5}{3}\right)$ ,  $f$  is increasing on the interval  $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$ .

Since  $f'(x) < 0$  on the interval  $3 \ln\left(\frac{5}{3}\right) < x < 4$ ,  $f$  is decreasing on the interval  $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$ .

Therefore,  $f$  has an absolute maximum at  $x = 3 \ln\left(\frac{5}{3}\right)$ .

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3. Refer to the graph and the information below.



The function  $f$  is defined on the closed interval  $[0, 8]$ . The graph of its derivative  $f'$  is shown above.

At what value of  $x$  does the absolute minimum of  $f$  occur?

- (A) 0  
(B) 2  
(C) 4  
(D) 6  
(E) 8



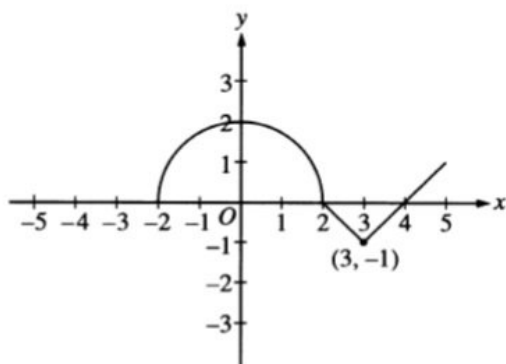
4.  Let  $f(x) = \int_{-2}^{x^2-3x} e^{t^2} dt$ . At what value of  $x$  is  $f(x)$  a minimum?

- (A) For no value of  $x$   
(B)  $\frac{1}{2}$   
(C)  $\frac{3}{2}$   
(D) 2  
(E) 3



5-4

5.



The graph of the function  $f$  consists of a semicircle and two line segments as shown above. Let  $g$  be the function given by  $g(x) = \int_0^x f(t) dt$ .

- Find  $g(3)$ .
- Find all the values of  $x$  on the open interval  $(-2, 5)$  at which  $g$  has a relative maximum. Justify your answer.
- Write an equation for the line tangent to the graph of  $g$  at  $x = 3$ .
- Find the  $x$ -coordinate of each point of inflection of the graph of  $g$  on the open interval  $(-2, 5)$ . Justify your answer.

**Part A**

2 points are earned for the correct answer

<-1> each incorrect area

<-1> error in summing

$$\begin{aligned}
 g(3) &= \int_0^3 f(t) dt \\
 &= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2}
 \end{aligned}$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

2 points are earned for the correct answer

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<-1> error in summing

5-4

$$g(3) = \int_0^3 f(t) dt$$

$$= \frac{1}{4} \cdot \pi \cdot 2^2 - \frac{1}{2} = \pi - \frac{1}{2}$$

**Part B**

1 point is earned for correctly answering that relative maximum at  $x = 2$  only

1 point is earned for correctly answering  $g'(x) = f(x)$  or interprets  $g(x)$  as area accumulator

1 point is earned for the correct justification (ignore discussion at  $x = 5$ )

$g(x)$  has relative maximum at  $x = 2$  because  $g'(x) = f(x)$  changes from positive to negative at  $x = 2$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

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**Part C**

1 point is earned for the correct answer  $g'(3) = -1$

$$g(3) = \pi - \frac{1}{2}$$

$$g'(3) = f(3) = -1$$

1 point is earned for the correct equation using  $g(3)$  and  $g'(3)$

$$y - \left( \pi - \frac{1}{2} \right) = -1(x - 3)$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

## 5-4

The student response earns all of the following points:

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1 point is earned for the correct equation using  $g(3)$  and  $g'(3)$

$$y - \left( \pi - \frac{1}{2} \right) = -1(x - 3)$$

**Part D**

1 point is earned for the correct points of inflection with  $x$ -coordinates 0 and 3 only

1 point is earned for the correct justification (ignore discussion at  $x = 2$ )

1/2 if  $x = 0, 3$  selected as candidates and  $x = 3$  discarded because  $g''(3)$  does not exist

1/2 if  $x = 0, 2, 3$  selected as candidates and  $x = 2$  and  $x = 3$  discarded because  $g''(2)$  and  $g''(3)$  do not exist

graph of  $g$  has points of inflection with  $x$ -coordinates  $x = 0$  and  $x = 3$

because  $g''(x) = f(x)$  changes from positive to negative at  $x = 0$  and from negative to positive at  $x = 3$

or

because  $g'(x) = f(x)$  changes from increasing to decreasing at  $x = 0$  and from decreasing to increasing at  $x = 3$



|   |   |   |
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| 0 | 1 | 2 |
|---|---|---|

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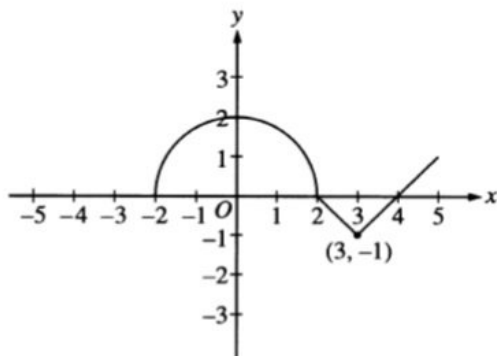
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or

because  $g'(x) = f(x)$  changes from increasing to decreasing at  $x = 0$  and from decreasing to increasing at  $x = 3$

6.



The graph of the function  $f$  consists of a semicircle and two line segments as shown above. Let  $g$  be the function given by  $g(x) = \int_0^x f(t) dt$ .

Find all the values of  $x$  on the open interval  $(-2, 5)$  at which  $g$  has a relative maximum. Justify your answer.

**Part B**

1 point is earned for correctly answering that relative maximum at  $x = 2$  only

1 point is earned for correctly answering  $g'(x) = f(x)$  or interprets  $g(x)$  as area accumulator

1 point is earned for the correct justification (ignore discussion at  $x = 5$ )

$g(x)$  has relative maximum at  $x = 2$  because  $g'(x) = f(x)$  changes from positive to negative at  $x = 2$



|   |   |   |   |
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| 0 | 1 | 2 | 3 |
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The student response earns three of the following points:

1 point is earned for correctly answering that relative maximum at  $x = 2$  only

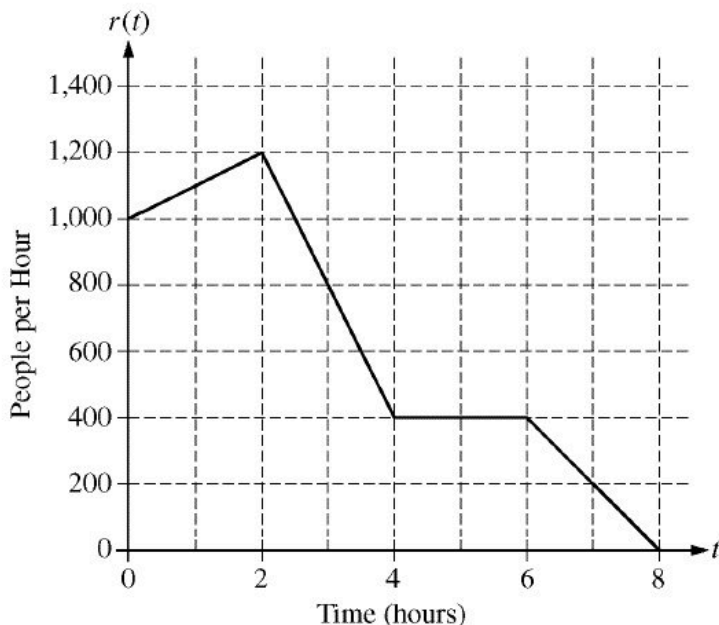
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1 point is earned for the correct justification (ignore discussion at  $x = 5$ )

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7.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate,  $r(t)$ , at which people arrive at the ride throughout the day. Time  $t$  is measured in hours from the time the ride begins operation.

At what time  $t$  is the line for the ride the longest? How many people are in line at that time? Justify your answers.

**Part C**

1 point is earned for identifying  $t=3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t)=800$  only at  $t=3$

For  $0 \leq t < 3$ ,  $r(t) > 800$ . For  $3 < t \leq 8$ ,  $r(t) < 800$ .

Therefore, the line is longest at time  $t=3$ .

There are  $700+3200-800 \cdot 3=1500$  people waiting in line at time  $t=3$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
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The student response earns all of the following points:

**5-4**

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$r(t)=800$  only at  $t=3$

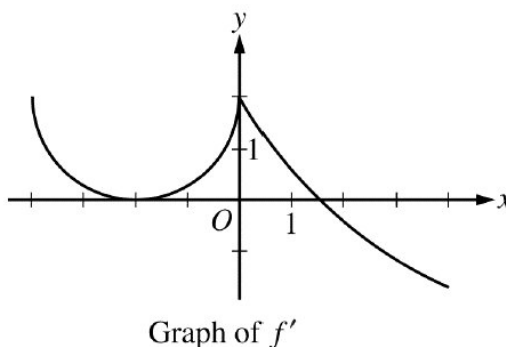
*For  $0 \leq t < 3$ ,  $r(t) > 800$ . For  $3 < t \leq 8$ ,  $r(t) < 800$ .*

Therefore, the line is longest at time  $t=3$ .

There are  $700+3200-800 \cdot 3=1500$  people waiting in line at time  $t=3$ .

5-4

8.



The derivative of a function  $f$  is defined by  $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$ .

The graph of the continuous function  $f$ , shown in the figure above, has  $x$ -intercepts at  $x = -2$  and  $x = 3 \ln\left(\frac{5}{3}\right)$ . The graph of  $g$  on  $-4 \leq x \leq 0$  is a semicircle, and  $f(0) = 5$ .

- (a) For  $-4 < x < 4$ , find all values of  $x$  at which the graph of  $f$  has a point of inflection. Justify your answer.  
 (b) Find  $f(-4)$  and  $f(4)$ .  
 (c) For  $-4 \leq x \leq 4$ , find the value of  $x$  at which  $f$  has an absolute maximum. Justify your answer.

**Part A**

1 point is earned if identifies  $x = -2$  or  $x = 0$

1 point is earned for answer with justification

$f$  changes from decreasing to increasing at  $x = -2$  and from increasing to decreasing at  $x = 0$ . Therefore, the graph of  $f$  has points of inflections at  $x = -2$  and  $x = 0$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
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1 point is earned if identifies  $x = -2$  or  $x = 0$

1 point is earned for answer with justification

$f$  changes from decreasing to increasing at  $x = -2$  and from increasing to decreasing at  $x = 0$ . Therefore, the graph of  $f$  has points of inflections at  $x = -2$  and  $x = 0$ .

**Part B**

1 point is earned for integral of  $f(-4)$

5-4

1 point is earned for value of  $f(-4)$ 1 point is earned for integral of  $f(4)$ 1 point is earned for antiderivative of  $f(4)$ 1 point is earned for value of  $f(4)$ 

$$\begin{aligned}
 f(-4) &= 5 + \int_0^{-4} g(x) dx \\
 &= 5 - (8 - 2\pi) = 2\pi - 3
 \end{aligned}$$

$$\begin{aligned}
 f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\
 &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\
 &= 8 - 15e^{-4/3}
 \end{aligned}$$



|   |   |   |   |   |   |
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| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|

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 &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\
 &= 8 - 15e^{-4/3}
 \end{aligned}$$

5-4

**Part C**

1 point is earned for the answer

1 point is earned for the justification

Since  $f'(x) > 0$  on the intervals  $-4 < x < -2$  and  $-2 < x < 3 \ln\left(\frac{5}{3}\right)$ ,  $f$  is increasing on the interval  $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$ .

Since  $f'(x) < 0$  on the interval  $3 \ln\left(\frac{5}{3}\right) < x < 4$ ,  $f$  is decreasing on the interval  $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$ .

Therefore,  $f$  has an absolute maximum at  $x = 3 \ln\left(\frac{5}{3}\right)$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

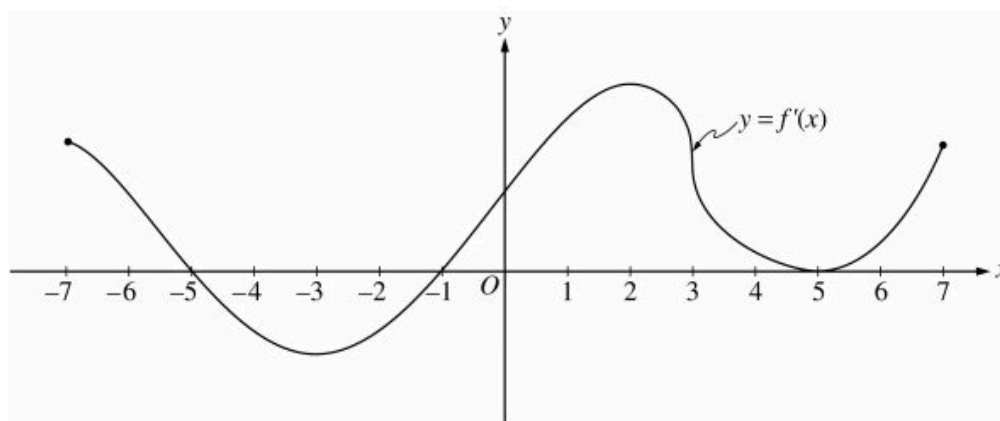
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Since  $f'(x) < 0$  on the interval  $3 \ln\left(\frac{5}{3}\right) < x < 4$ ,  $f$  is decreasing on the interval  $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$ .

Therefore,  $f$  has an absolute maximum at  $x = 3 \ln\left(\frac{5}{3}\right)$ .

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9.



The figure above shows the graph of  $f'$ , the derivative of the function  $f$ , for  $-7 \leq x \leq 7$ . The graph of  $f$  has horizontal tangent lines at  $x = -3$ ,  $x = 2$ , and  $x = 5$ , and a vertical tangent line at  $x = 3$ .

- Find all values of  $x$ , for  $-7 < x < 7$ , at which  $f$  attains a relative minimum. Justify your answer.
- Find all values of  $x$ , for  $-7 < x < 7$ , at which  $f$  attains a relative maximum. Justify your answer.
- Find all values of  $x$ , for  $-7 < x < 7$ , at which  $f''(x) < 0$ .
- At what value of  $x$ , for  $-7 \leq x \leq 7$ , does  $f$  attain its absolute maximum? Justify your answer.

**Part A**

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$  changes from negative to positive at  $x = -1$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
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The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$  changes from negative to positive at  $x = -1$

**Part B**

1 point is earned for the answer

**5-4**

1 point is earned for the justification

$$x = -5$$

$f'(x)$  changes from positive to negative at  $x = -5$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$x = -5$$

$f'(x)$  changes from positive to negative at  $x = -5$

**Part C**

1 point is earned for  $(-7, -3)$

1 point is earned for  $(2, 3) \cup (3, 5)$

$f''(x)$  exists and  $f'$  is decreasing on the intervals  $(-7, -3)$ ,  $(2, 3)$ , and  $(3, 5)$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $(-7, -3)$

1 point is earned for  $(2, 3) \cup (3, 5)$

$f''(x)$  exists and  $f'$  is decreasing on the intervals  $(-7, -3)$ ,  $(2, 3)$ , and  $(3, 5)$

**Part D**

1 point is earned for the answer

1 point is earned for identifying  $x = -5$  and  $x = 7$  as candidates

- or -

indicating that the graph of  $f$  increases, decreases, then increases

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1 point is earned for justifying  $f(7) > f(-5)$

$$x = 7$$

The absolute maximum must occur at  $x = -5$  or at an endpoint.

$f(-5) > f(-7)$  because  $f$  is increasing on  $(-7, -5)$

The graph of  $f'$  shows that the magnitude of the negative change in  $f$  from  $x = -5$  to  $x = -1$  is smaller than the positive change in  $f$  from  $x = -1$  to  $x = 7$ . Therefore the net change in  $f$  is positive from  $x = -5$  to  $x = 7$ , and  $f(7) > f(-5)$ . So  $f(7)$  is the absolute maximum.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

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- or -

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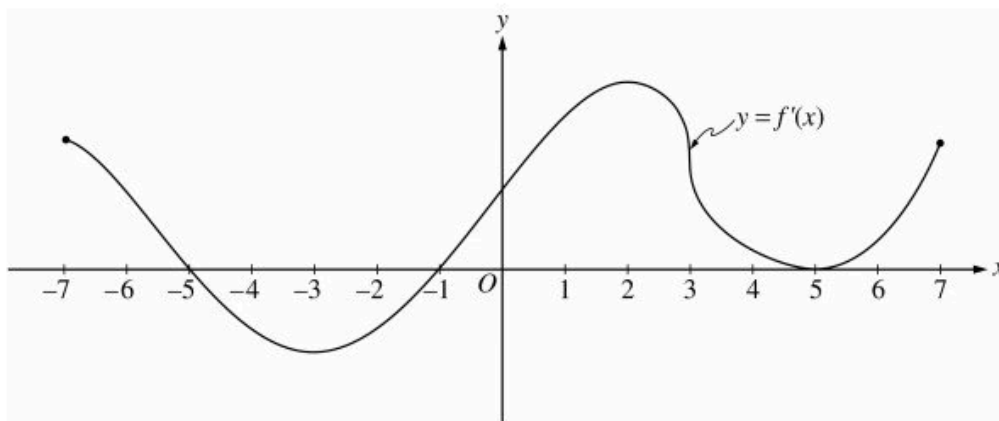
$$x = 7$$

The absolute maximum must occur at  $x = -5$  or at an endpoint.

$f(-5) > f(-7)$  because  $f$  is increasing on  $(-7, -5)$

The graph of  $f'$  shows that the magnitude of the negative change in  $f$  from  $x = -5$  to  $x = -1$  is smaller than the positive change in  $f$  from  $x = -1$  to  $x = 7$ . Therefore the net change in  $f$  is positive from  $x = -5$  to  $x = 7$ , and  $f(7) > f(-5)$ . So  $f(7)$  is the absolute maximum.

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The figure above shows the graph of  $f'$ , the derivative of the function  $f$ , for  $-7 \leq x \leq 7$ . The graph of  $f$  has horizontal tangent lines at  $x = -3$ ,  $x = 2$ , and  $x = 5$ , and a vertical tangent line at  $x = 3$ .

10. Find all values of  $x$ , for  $-7 < x < 7$ , at which  $f$  attains a relative minimum. Justify your answer.

**Part A**

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$  changes from negative to positive at  $x = -1$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$  changes from negative to positive at  $x = -1$

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11. Find all values of  $x$ , for  $-7 < x < 7$ , at which  $f$  attains a relative maximum. Justify your answer.
- 

**Part B**

1 point is earned for the answer

1 point is earned for the justification

$x = -5$

$f'(x)$  changes from positive to negative at  $x = -5$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
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The student response earns all of the following points:

1 point is earned for the answer

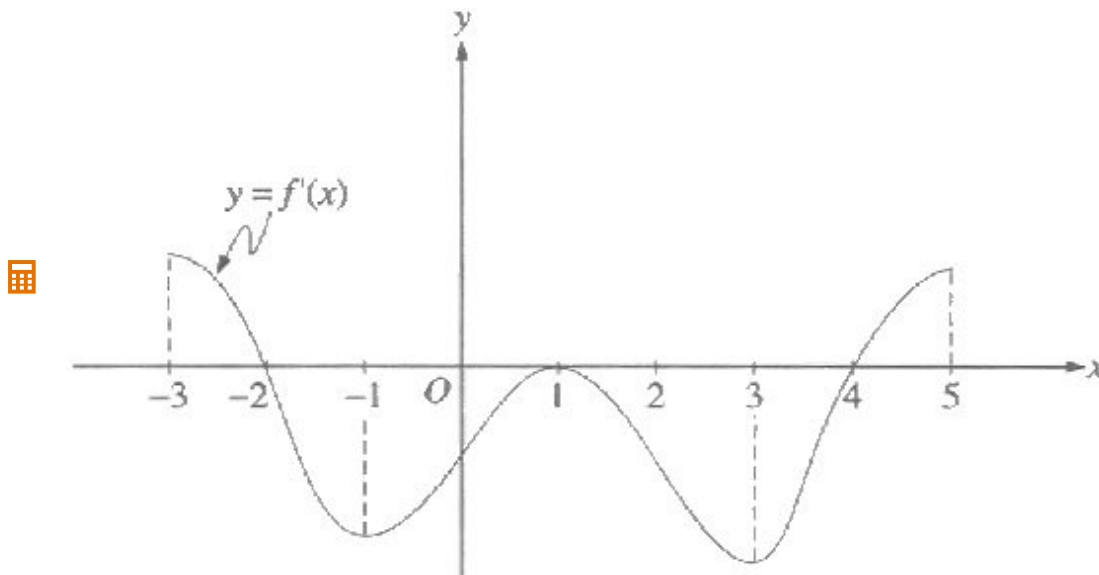
1 point is earned for the justification

$x = -5$

$f'(x)$  changes from positive to negative at  $x = -5$

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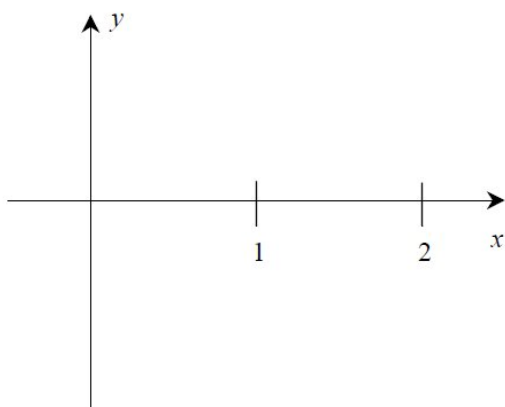
12.



Note: This is the graph of the derivative of  $f$ , not the graph of  $f$ .

The figure above shows the graph of  $f'$ , the derivative of a function  $f$ . The domain of  $f$  is the set of all real numbers  $x$  such that  $-3 < x < 5$ .

- For what values of  $x$  does  $f$  have a relative maximum? Why?
- For what values of  $x$  does  $f$  have a relative minimum? Why?
- On what intervals is the graph of  $f$  concave upward? Use  $f'$  to justify your answer.
- Suppose that  $f(1) = 0$ . In the  $xy$ -plane provided, draw a sketch that shows the general shape of the graph of the function  $f$  on the open interval  $0 < x < 2$ .



### Part A

1 point is earned for the correct answer  $x = -2$

$x = -2$

1 point is earned for the correct reason

$f'(x)$  changes from positive to negative at  $x = -2$  OR  $f$  is increasing to the left of  $x = -2$  and decreasing to the right of  $x = -2$ . OR

**5-4**

$$f'(-2) = 0 \text{ and } f''(-2) < 0$$

0/2 for point other than  $-2$  or  $5$ 

3/4 points between Part A and Part B if maximum and minimum are reversed



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct answer  $x = -2$ 

1 point is earned for the correct reason

$f'(x)$  changes from positive to negative at  $x = -2$  OR  $f$  is increasing to the left of  $x = -2$  and decreasing to the right of  $x = -2$ .

$$\text{OR } f'(-2) = 0 \text{ and } f''(-2) < 0$$

0/2 for point other than  $-2$  or  $4$ 

3/4 points between Part A and Part B if reverse maximum and minimum

**Part B**1 point is earned for the correct answer  $x = 4$ 

$$x = 4$$

1 point is earned for the correct reason

$f'(x)$  changes from negative to positive at  $x = 4$  OR  $f$  is decreasing to the left of  $x = 4$  and increasing to the right of  $x = 4$ .

3/4 points between Part A and Part B if reverse maximum and minimum

&lt;-1&gt; for mismatching if clear



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct answer  $x = 4$ 

$$x = 4$$

**5-4**

1 point is earned for the correct reason

$f(x)$  changes from negative to positive at  $x = 4$  OR  $f$  is decreasing to the left of  $x = 4$  and increasing to the right of  $x = 4$ .

3/4 points between Part A and Part B if reverse maximum and minimum

<-1> for mismatching if clear

**Part C**

1 point is earned for the correct interval  $(-1, 1)$

1 point is earned for the correct interval  $(3, 5)$

$(-1, 1)$  and  $(3, 5)$

1 point is earned for the correct reason using  $f'$

$f'$  is increasing on these intervals

Max 1/3 for one incorrect interval

0/3 for two incorrect intervals



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct interval  $(-1, 1)$

1 point is earned for the correct interval  $(3, 5)$

$(-1, 1)$  and  $(3, 5)$

1 point is earned for the correct reason using  $f'$

$f'$  is increasing on these intervals

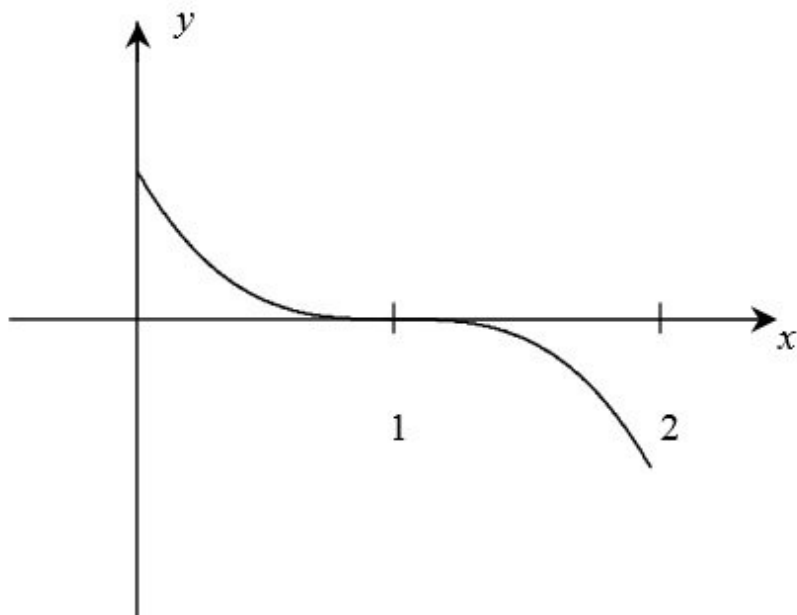
Max 1/3 for one incorrect interval

0/3 for two incorrect intervals

**Part D**

2 points are earned for the correct graph

5-4

<-1>  $f(1) \neq 0$ 

&lt;-1&gt; not decreasing

&lt;-1&gt; incorrect concavity

<-1>  $f'(1) \neq 0$ 

Incorrect use of standard notation deduct point only once

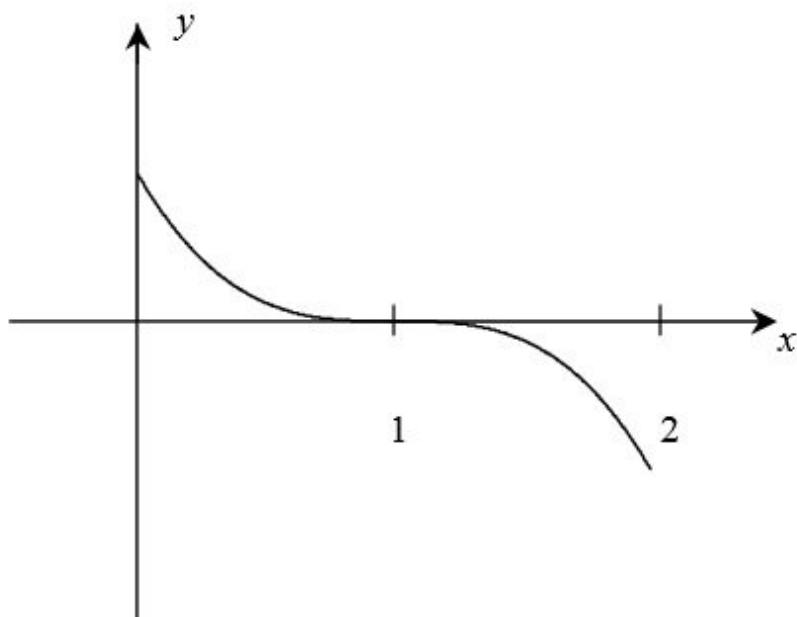


|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

2 points are earned for the correct graph

5-4

<-1>  $f(1) \neq 0$ 

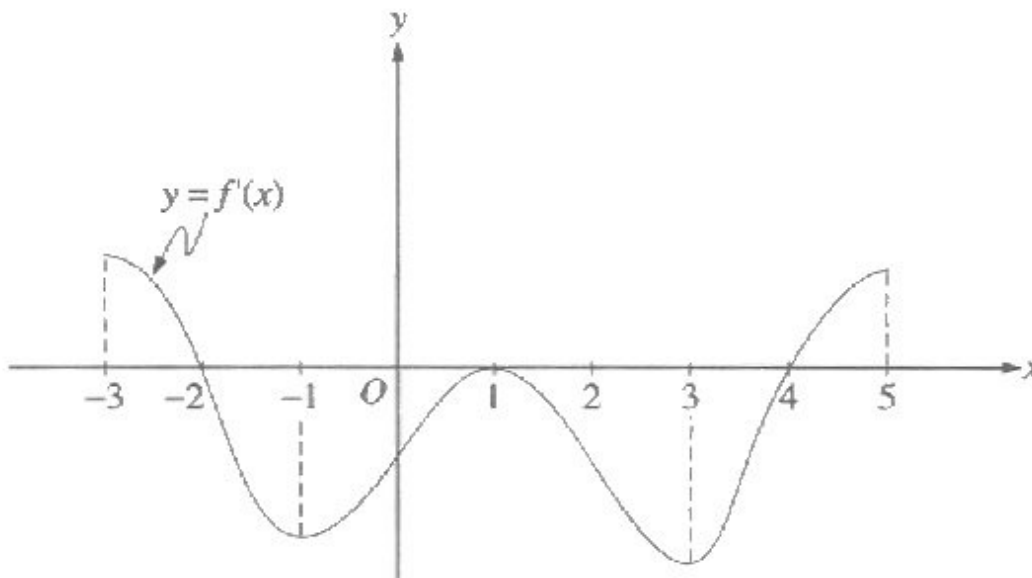
&lt;-1&gt; not decreasing

&lt;-1&gt; incorrect concavity

<-1>  $f'(1) \neq 0$ 

Incorrect use of standard notation deduct point only once

5-4



Note: This is the graph of the derivative of  $f$ , not the graph of  $f$ .

The figure above shows the graph of  $f'$ , the derivative of a function  $f$ . The domain of  $f$  is the set of all real numbers  $x$  such that  $-3 < x < 5$ .

13. For what values of  $x$  does  $f$  have a relative maximum? Why?

**Part A**

1 point is earned for the correct answer  $x = -2$  or  $5$

$x = -2$

1 point is earned for the correct reason

$f'(x)$  changes from positive to negative at  $x = -2$  OR  $f$  is increasing to the left of  $x = -2$  and decreasing to the right of  $x = -2$ .

$f'(-2) = 0$  and  $f''(-2) < 0$

2/2 if  $x = 5$  is justified

< -1 > if  $x = 5$  is not justified

0/2 for point other than -2 or 5

3/4 points between Part A and Part B if reverse maximum and minimum

< -1 > for mismatching if clear

5-4



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for the correct answer  $x = -2$  or 5

$$x = -2$$

1 point is earned for the correct reason

$f'(x)$  changes from positive to negative at  $x = -2$  OR  $f$  is increasing to the left of  $x = -2$  and decreasing to the right of  $x = -2$ .

$$f'(-2) = 0 \text{ and } f''(-2) < 0$$

2/2 if  $x = 5$  is justified

< -1 > if  $x = 5$  is not justified

0/2 for point other than -2 or 5

3/4 points between Part A and Part B if reverse maximum and minimum

< -1 > for mismatching if clear

5-4

14. For what values of  $x$  does  $f$  have a relative minimum? Why?

---

**Part B**

1 point is earned for the correct answer  $x = 4$

$x = 4$

1 point is earned for the correct reason

$f'(x)$  changes from negative to positive at  $x = 4$  OR  $f$  is decreasing to the left of  $x = 4$  and increasing to the right of  $x = 4$ .

3/4 points between Part A and Part B if reverse maximum and minimum

< -1 > for mismatching if clear



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for the correct answer  $x = 4$

$x = 4$

1 point is earned for the correct reason

$f'(x)$  changes from negative to positive at  $x = 4$  OR  $f$  is decreasing to the left of  $x = 4$  and increasing to the right of  $x = 4$ .

3/4 points between Part A and Part B if reverse maximum and minimum

< -1 > for mismatching if clear

## 5-4

15. Let  $h$  be a function defined for all  $x \neq 0$  such that  $h(4) = -3$  and the derivative of  $h$  is given by  $h'(x) = \frac{x^2 - 2}{x}$  for all  $x \neq 0$ .
- (a) Find all values of  $x$  for which the graph of  $h$  has a horizontal tangent, and determine whether  $h$  has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (b) On what intervals, if any, is the graph of  $h$  concave up? Justify your answer.
- (c) Write an equation for the line tangent to the graph of  $h$  at  $x = 4$ .
- (d) Does the line tangent to the graph of  $h$  at  $x = 4$  lie above or below the graph of  $h$  for  $x > 4$ ? Why?

## Part A

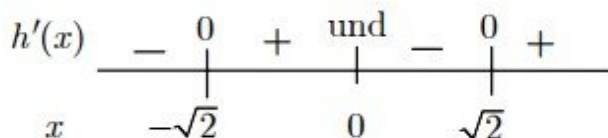
1 point is earned for:  $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at  $x = -\sqrt{2}$  and at  $x = \sqrt{2}$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $x = \pm\sqrt{2}$

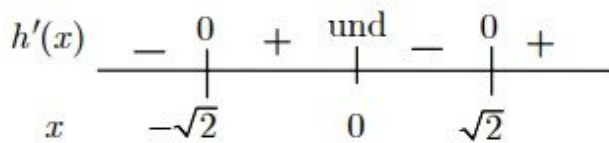
1 point is earned for the analysis

2 points are earned for the conclusions

5-4

&lt;-1&gt; not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$

Local minima at  $x = -\sqrt{2}$  and at  $x = \sqrt{2}$ **Part B**1 point is earned for:  $h''(x)$ 1 point is earned for:  $h''(x) > 0$ 

1 point is earned for the answer

$$h''(x) = 1 + \frac{2}{x^2} > 0 \text{ for all } x \neq 0. \text{ Therefore, the graph of } h \text{ is concave up for all } x \neq 0.$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $h''(x)$ 1 point is earned for:  $h''(x) > 0$ 

1 point is earned for the answer

$$h''(x) = 1 + \frac{2}{x^2} > 0 \text{ for all } x \neq 0. \text{ Therefore, the graph of } h \text{ is concave up for all } x \neq 0.$$

**Part C**

1 point is earned for the tangent line equation

5-4

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$

**Part D**

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of  $h$  is concave up for  $x > 4$ .



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of  $h$  is concave up for  $x > 4$ .

5-4

16. Let  $h$  be a function defined for all  $x \neq 0$  such that  $h(4) = -3$  and the derivative of  $h$  is given by

$$h'(x) = \frac{x^2 - 2}{x} \text{ for all } x \neq 0.$$

Find all values of  $x$  for which the graph of  $h$  has a horizontal tangent, and determine whether  $h$  has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

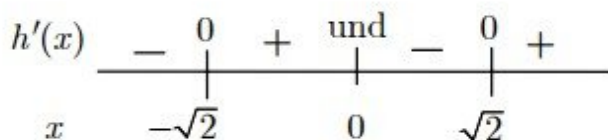
**Part A**

1 point is earned for:  $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions < -1 > not dealing with discontinuity at 0

$$h'(x) = 0, \text{ at } x = \pm\sqrt{2}$$



Local minima at  $x = -\sqrt{2}$  and at  $x = \sqrt{2}$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

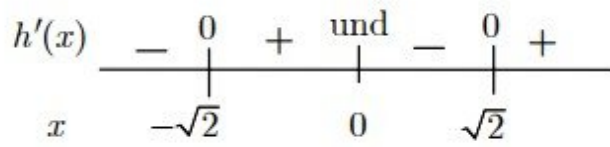
1 point is earned for:  $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions < -1 > not dealing with discontinuity at 0

$$h'(x) = 0, \text{ at } x = \pm\sqrt{2}$$

5-4



Local minima at  $x = -\sqrt{2}$  and at  $x = \sqrt{2}$

5-4

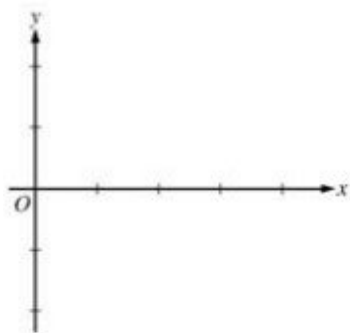
17.

| $x$      | 0  | $0 < x < 1$ | 1 | $1 < x < 2$ | 2   | $2 < x < 3$ | 3  | $3 < x < 4$ |
|----------|----|-------------|---|-------------|-----|-------------|----|-------------|
| $f(x)$   | -1 | Negative    | 0 | Positive    | 2   | Positive    | 0  | Negative    |
| $f'(x)$  | 4  | Positive    | 0 | Positive    | DNE | Negative    | -3 | Negative    |
| $f''(x)$ | -2 | Negative    | 0 | Positive    | DNE | Negative    | 0  | Positive    |

Let  $f$  be a function that is continuous on the interval  $[0, 4)$ . The function  $f$  is twice differentiable except at  $x = 2$ . The function  $f$  and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of  $f$  do not exist at  $x = 2$ .

(a) For  $0 < x < 4$ , find all values of  $x$  at which  $f$  has a relative extremum. Determine whether  $f$  has a relative maximum or a relative minimum at each of these values. Justify your answer.

(b) On the axes provided, sketch the graph of a function that has all the characteristics of  $f$ .



(c) Let  $g$  be the function defined by  $g(x) = \int_1^x f(t)dt$  on the open interval  $(0, 4)$ . For  $0 < x < 4$ , find all values of  $x$  at which  $g$  has a relative extremum. Determine whether  $g$  has a relative maximum or a relative minimum at each of these values. Justify your answer.

(d) For the function  $g$  defined in part (c), find all values of  $x$ , for  $0 < x < 4$ , at which the graph of  $g$  has a point of inflection. Justify your answer.

### Part A

1 point is earned for the relative extremum at  $x = 2$

1 point is earned for the relative maximum with justification

$f$  has a relative maximum at  $x = 2$  because  $f'$  changes from positive to negative at  $x = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the relative extremum at  $x = 2$

5-4

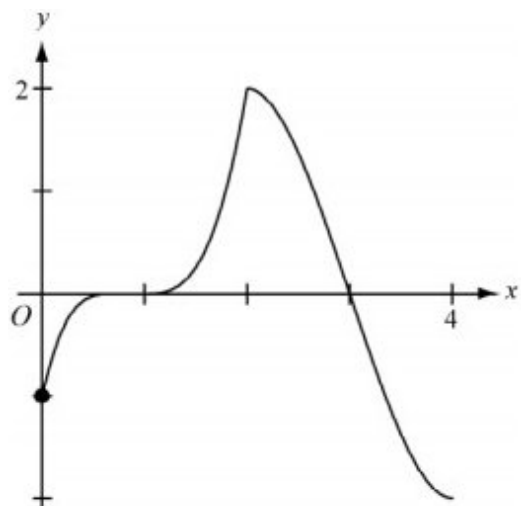
1 point is earned for the relative maximum with justification

$f$  has a relative maximum at  $x = 2$  because  $f'$  changes from positive to negative at  $x = 2$ .

**Part B**

1 point is earned for points at  $x = 0, 1, 2, 3$  and behavior at  $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior



✓

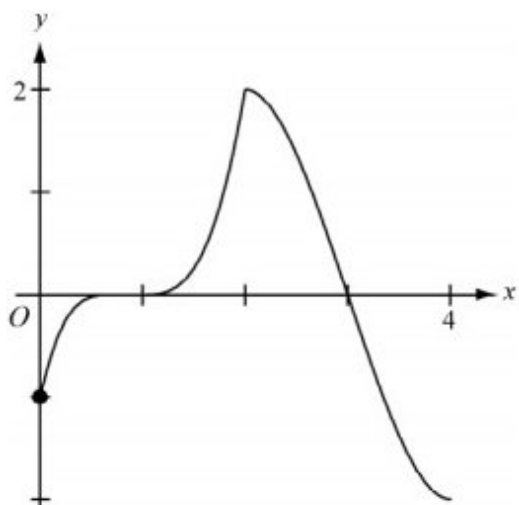
|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for points at  $x = 0, 1, 2, 3$  and behavior at  $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior

5-4

**Part C**

1 point is earned for:  $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$  at  $x = 1, 3$ .

$g'$  changes from negative to positive at  $x = 1$  so  $g$  has a relative minimum at  $x = 1$ .  $g'$  changes from positive to negative at  $x = 3$  so  $g$  has a relative maximum at  $x = 3$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$  at  $x = 1, 3$ .

$g'$  changes from negative to positive at  $x = 1$  so  $g$  has a relative minimum at  $x = 1$ .  $g'$  changes from positive to negative at  $x = 3$  so  $g$  has a relative maximum at  $x = 3$ .

**Part D**

1 point is earned for:  $x = 2$

5-4

1 point is earned for the answer with justification

The graph of  $g$  has a point of inflection at  $x = 2$  because  $g'' = f'$  changes sign at  $x = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $x = 2$

1 point is earned for the answer with justification

The graph of  $g$  has a point of inflection at  $x = 2$  because  $g'' = f'$  changes sign at  $x = 2$ .

| $x$      | 0  | $0 < x < 1$ | 1 | $1 < x < 2$ | 2   | $2 < x < 3$ | 3  | $3 < x < 4$ |
|----------|----|-------------|---|-------------|-----|-------------|----|-------------|
| $f(x)$   | -1 | Negative    | 0 | Positive    | 2   | Positive    | 0  | Negative    |
| $f'(x)$  | 4  | Positive    | 0 | Positive    | DNE | Negative    | -3 | Negative    |
| $f''(x)$ | -2 | Negative    | 0 | Positive    | DNE | Negative    | 0  | Positive    |

Let  $f$  be a function that is continuous on the interval  $[0, 4)$ . The function  $f$  is twice differentiable except at  $x = 2$ . The function  $f$  and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of  $f$  do not exist at  $x = 2$ .

18. For  $0 < x < 4$ , find all values of  $x$  at which  $f$  has a relative extremum. Determine whether  $f$  has a maximum or a relative minimum at each of these values. Justify your answer.

### Part A

1 point is earned for the relative extremum at  $x = 2$

1 point is earned for the relative maximum with justification

$f$  has a relative maximum at  $x = 2$  because  $f'$  changes from positive to negative at  $x = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the relative extremum at  $x = 2$

1 point is earned for the relative maximum with justification

$f$  has a relative maximum at  $x = 2$  because  $f'$  changes from positive to negative at  $x = 2$ .

5-4

19. Let  $g$  be the function defined by  $g(x) = \int_1^x f(t)dt$  on the open interval  $(0, 4)$ . For  $0 < x < 4$ , find all values of  $x$  at which  $g$  has a relative extremum. Determine whether  $g$  has a relative maximum or a relative minimum at each of these values. Justify your answer.

**Part C**

1 point is earned for:  $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$  at  $x = 1, 3$ .

$g'$  changes from negative to positive at  $x = 1$  so  $g$  has a relative minimum at  $x = 1$ .  $g'$  changes from positive to negative at  $x = 3$  so  $g$  has a relative maximum at  $x = 3$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $g'(x) = f(x)$

1 point is earned for the critical points

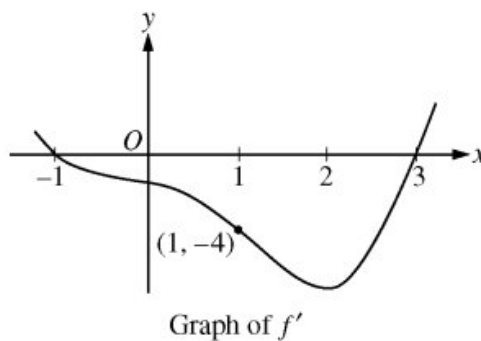
1 point is earned for the answer with justification

$g'(x) = f(x) = 0$  at  $x = 1, 3$ .

$g'$  changes from negative to positive at  $x = 1$  so  $g$  has a relative minimum at  $x = 1$ .  $g'$  changes from positive to negative at  $x = 3$  so  $g$  has a relative maximum at  $x = 3$ .

5-4

20.



Let  $f$  be a twice-differentiable function defined on the interval  $-1.2 < x < 3.2$  with  $f(1) = 2$ . The graph of  $f'$ , the derivative of  $f$ , is shown above. The graph of  $f'$  crosses the  $x$ -axis at  $x = -1$  and  $x = 3$  and has a horizontal tangent at  $x = 2$ . Let  $g$  be the function given by  $g(x) = e^{f(x)}$ .

- Write an equation for the line tangent to the graph of  $g$  at  $x = 1$ .
- For  $-1.2 < x < 3.2$ , find all values of  $x$  at which  $g$  has a local maximum. Justify your answer.
- The second derivative of  $g$  is  $g''(x) = e^{f(x)} [(f'(x))^2 + f''(x)]$ . Is  $g''(-1)$  positive, negative, or zero? Justify your answer.
- Find the average rate of change of  $g'$ , the derivative of  $g$ , over the interval  $[1, 3]$ .

**Part A**

1 point is earned for  $g'(x)$

1 point is earned for  $g(1)$  and  $g'(1)$

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by  $y = e^2 - 4e^2(x - 1)$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $g'(x)$

1 point is earned for  $g(1)$  and  $g'(1)$

## 5-4

1 point is earned for tangent line equation

$$g(1) = e^{f(1)} = e^2$$

$$g'(x) = e^{f(x)} f'(x), \quad g'(1) = e^{f(1)} f'(1) = -4e^2$$

The tangent line is given by  $y = e^2 - 4e^2(x - 1)$ .

**Part B**

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So,  $g'$  changes from positive to negative only when  $f'$  changes from positive to negative. This occurs at  $x = -1$  only. Thus,  $g$  has a local maximum at  $x = -1$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So,  $g'$  changes from positive to negative only when  $f'$  changes from positive to negative. This occurs at  $x = -1$  only. Thus,  $g$  has a local maximum at  $x = -1$ .

**Part C**

## 5-4

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[ (f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since  $f'$  is decreasing on a neighborhood of  $-1$ ,  $f''(-1) < 0$ . Therefore,  $g''(-1) < 0$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$g''(-1) = e^{f(-1)} \left[ (f'(-1))^2 + f''(-1) \right]$$

$$e^{f(-1)} > 0 \text{ and } f'(-1) = 0$$

Since  $f'$  is decreasing on a neighborhood of  $-1$ ,  $f''(-1) < 0$ . Therefore,  $g''(-1) < 0$ .

**Part D**

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{e^{f(3)} f'(3) - e^{f(1)} f'(1)}{2} = 2e^2$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the difference quotient

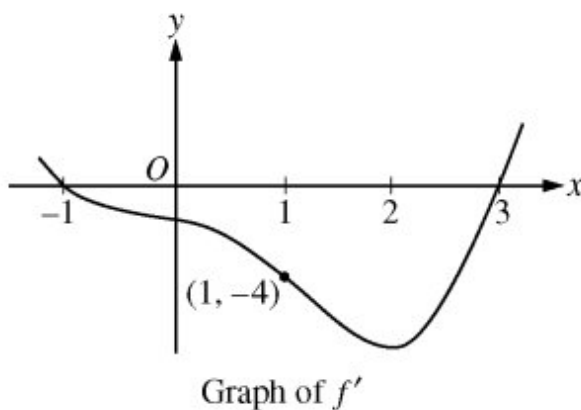
**5-4**

1 point is earned for the answer

$$\frac{g'(3) - g'(1)}{3 - 1} = \frac{ef(3)f'(3) - ef(1)f'(1)}{2} = 2e^2$$

5-4

21.



Let  $f$  be a twice-differentiable function defined on the interval  $-1.2 < x < 3.2$  with  $f(1) = 2$ . The graph of  $f'$ , the derivative of  $f$ , is shown above. The graph of  $f'$  crosses the  $x$ -axis at  $x = -1$  and  $x = 3$  and has a horizontal tangent at  $x = 2$ . Let  $g$  be the function given by  $g(x) = e^{f(x)}$ .

For  $-1.2 < x < 3.2$ , find all values of  $x$  at which  $g$  has a local maximum. Justify your answer.

**Part B**

1 point is earned for the answer

1 point is earned for the justification

$$g'(x) = e^{f(x)} f'(x)$$

$e^{f(x)} > 0$  for all  $x$ . So,  $g'$  changes from positive to negative only when  $f'$  changes from positive to negative. This occurs at  $x = -1$  only. Thus,  $g$  has a local maximum at  $x = -1$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

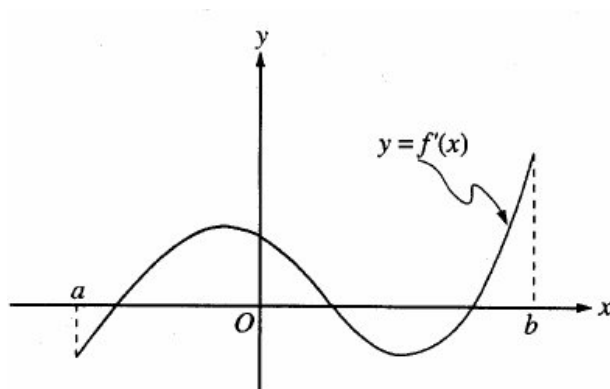
$$g'(x) = e^{f(x)} f'(x)$$

$$e^{f(x)} > 0 \text{ for all } x$$

So,  $g'$  changes from positive to negative only when  $f'$  changes from positive to negative. This occurs at  $x = -1$  only. Thus,  $g$  has a local maximum at  $x = -1$ .

5-4

22.



The graph of  $f'$ , the derivative of  $f$ , is shown in the figure above. Which of the following describes all relative extrema of  $f$  on the open interval  $(a, b)$ ?

- (A) One relative maximum and two relative minima
- (B) Two relative maxima and one relative minimum
- (C) Three relative maxima and one relative minimum
- (D) One relative maximum and three relative minima
- (E) Three relative maxima and two relative minima



## 5-4

23. Let  $f$  be the function given by  $f(x) = \ln|x/1 + x^2|$ .

(a) Find the domain of  $f$ .

(b) Determine whether  $f$  is an even function, an odd function, or neither. Justify your conclusion.

(c) At what values of  $x$  does  $f$  have a relative maximum or a relative minimum? For each such  $x$ , use the first derivative test to determine whether  $f(x)$  is a relative maximum or a relative minimum.

(d) Find the range of  $f$ .

**Part A**

1 point is earned for answer

$$x \neq 0$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for answer

$$x \neq 0$$

**Part B**

1 point is earned for the answer

1 point is earned for identification using  $f(x)$

Even

$$f(-x) = \ln \left| \frac{-x}{1 + (-x)^2} \right| = f(x)$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

5-4

1 point is earned for identification using  $f(x)$ 

Even

$$f(-x) = \ln \left| \frac{-x}{1 + (-x)^2} \right| = f(x)$$

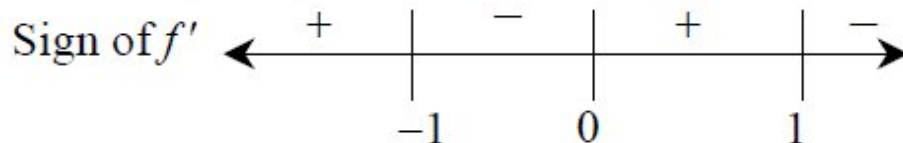
**Part C**

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of  $f'(x)$  and conclusion

$$\begin{aligned} f'(x) &= \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2} \\ &= \frac{1-x^2}{x(1+x^2)} \end{aligned}$$

 $f(x)$  has relative max at  $x = 1$  $f(x)$  has relative max at  $x = -1$ 

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

2 points are earned for the derivative

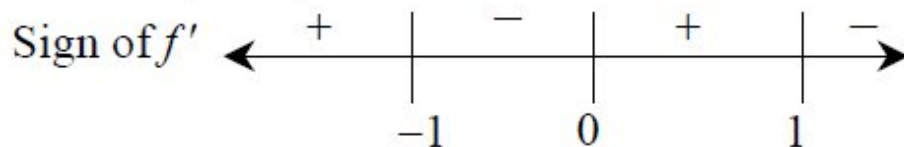
1 point is earned for finding critical number

1 point is earned for sign analysis of  $f'(x)$  and conclusion

5-4

$$f'(x) = \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$  has relative max at  $x = 1$

$f(x)$  has relative max at  $x = -1$

**Part D**

2 points are earned for the answer

1 point **deducted** for error

max is

$$f(1) = \ln \frac{1}{2} = -\ln 2$$

$$\lim_{x \rightarrow 0^+} f(x) = -\infty$$

$$\left( \text{or } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = -\infty \right)$$

$$\therefore \text{range} = \left( -\infty, \ln \frac{1}{2} \right]$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

2 points are earned for the answer

1 point **deducted** for error

max is

5-4

$$f(1) = \ln \frac{1}{2} = -\ln 2$$

$$\lim_{x \rightarrow 0^+} f(x) = -\infty$$

$$\left( \text{or } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = -\infty \right)$$

$$\therefore \text{range} = \left( -\infty, \ln \frac{1}{2} \right]$$

## 5-4

24. Let  $f$  be the function given by  $f(x) = \ln|x/1+x^2|$

At what values of  $x$  does  $f$  have a relative maximum or a relative minimum? For each such  $x$ , use the first derivative test to determine whether  $f(x)$  is a relative maximum or a relative minimum.

## Part C

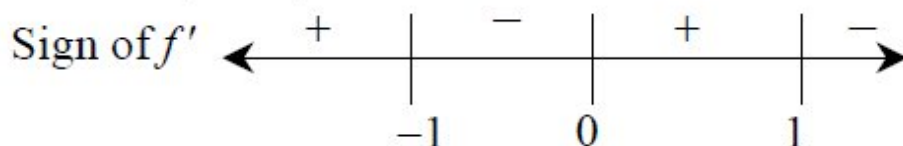
2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of  $f'(x)$  and conclusion

$$f'(-x) = \frac{1-x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$  has relative max at  $x = 1$

$f(x)$  has relative max at  $x = -1$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns four of the following points:

2 points are earned for the derivative

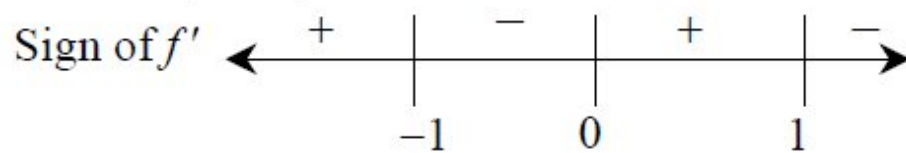
1 point is earned for finding critical number

1 point is earned for sign analysis of  $f'(x)$  and conclusion

$$f'(-x) = \frac{1-x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$

$$= \frac{1-x^2}{x(1+x^2)}$$

5-4

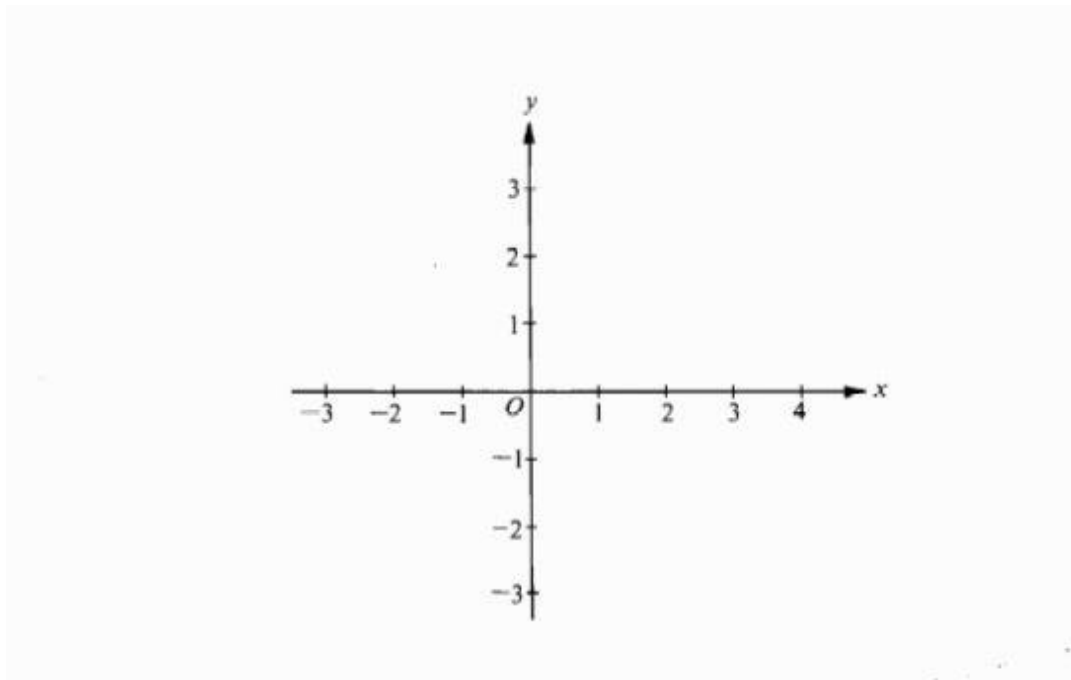


$f(x)$  has relative max at  $x = 1$

$f(x)$  has relative max at  $x = -1$

5-4

25. Let  $f$  be the function defined by  $f(x) = 2xe^{-x}$  for all real numbers  $x$ .
- (a) Write an equation of the horizontal asymptote for the graph of  $f$ .
  - (b) Find the  $x$ -coordinate of each critical point of  $f$ . For each such  $x$ , determine whether  $f(x)$  is a relative maximum, a relative minimum, or neither.
  - (c) For what values of  $x$  is the graph of  $f$  concave down?
  - (d) Using the results found in parts (a), (b), and (c), sketch the graph of  $y = f(x)$  in the  $xy$ -plane provided below.

**Part A**

1 point is earned for the correct equation

$$y = 0$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the correct equation

$$y = 0$$

**Part B**

1 point is earned for correct derivative

**5-4**1 point is earned for critical value for  $f'$ 

1 point is earned for identifying critical point as relative maximum

$$f'(x) = 2(-xe^{-x} + e^{-x})$$

$$= 2e^{-x}(1 - x)$$

critical point at  $x = 1$ relative maximum at  $x = 1$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for  $f'$ 

1 point is earned for identifying critical point as relative maximum

$$f'(x) = 2(-xe^{-x} + e^{-x})$$

$$= 2e^{-x}(1 - x)$$

critical point at  $x = 1$ relative maximum at  $x = 1$ **Part C**1 point is earned for correct  $f''(x)$  for  $f'(x)$  found in (b)

1 point is earned for correct interval

$$f''(x) = 2e^{-x}(-1) + (-2e^{-x})(1 - x)$$

$$= 2e^{-x}(x - 2)$$

Concave down when

5-4

$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for correct  $f''(x)$  for  $f'(x)$  found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

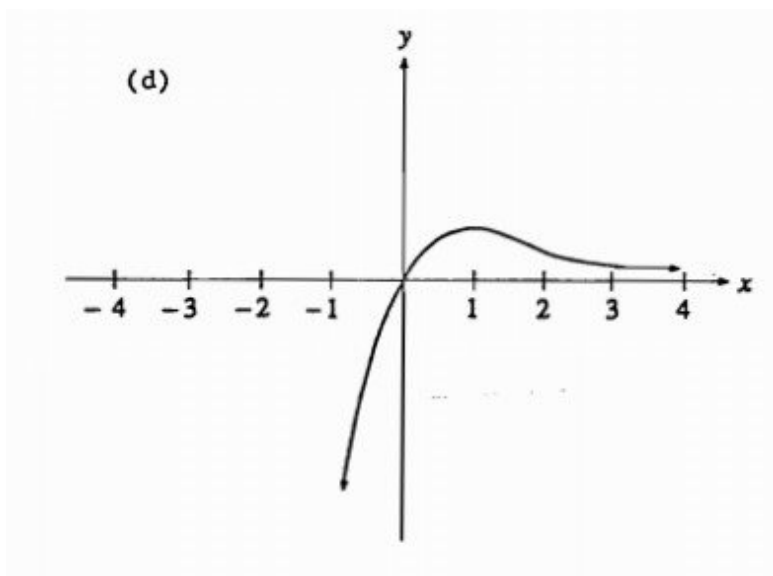
$$2e^{-x}(x-2) < 0$$

$$(x-2) < 0$$

$$x < 2$$

### Part D

3 points are earned for the graph consistent with information found in (a), (b), and (c)



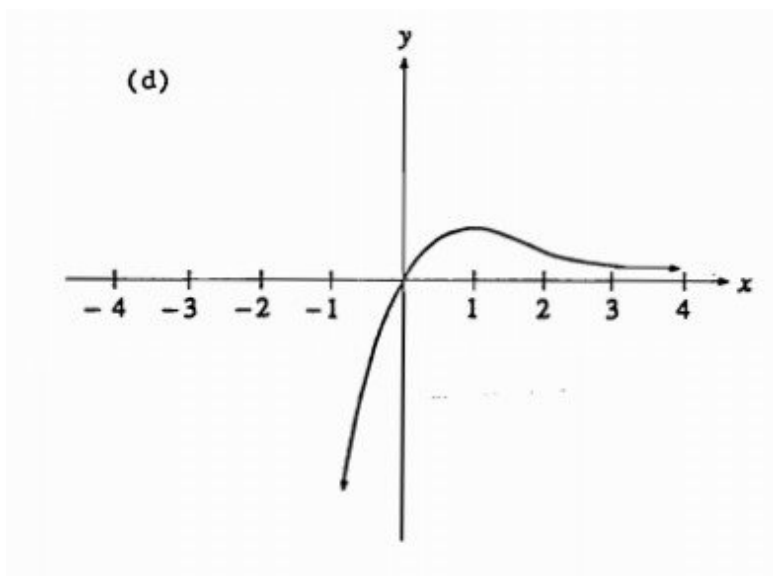
5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

3 points are earned for the graph consistent with information found in (a), (b), and (c)



5-4

26. Let  $F$  be the function defined by  $f(x) = 2xe^{-x}$  for all real numbers  $x$ .

Find the  $x$ -coordinate of each critical point of  $F$ . For each such  $x$ , determine whether  $f(x)$  is a relative maximum, a relative minimum, or neither.

**Part B**

1 point is earned for correct derivative

1 point is earned for critical value for  $f'$

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at  $x=1$

relative maximum at  $x=1$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for  $f'$

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at  $x=1$

relative maximum at  $x=1$

## 5-4

27. A cubic polynomial function  $f$  is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where  $a$ ,  $b$ , and  $k$  are constants. The function  $f$  has a local minimum at  $x = -1$ , and the graph  $f$  has a point of inflection at  $x = -2$ .

(a) Find the values of  $a$  and  $b$ .

(b) If  $\int_0^1 f(x) dx = 32$ , what is the value of  $k$ ?

**Part A**

1 point is earned for:  $f'(x)$

1 point is earned for:  $f''(x)$

1 point is earned for:  $f'(-1) = 0$

1 point is earned for:  $f''(-2) = 0$

1 point is earned for:  $a$ ,  $b$

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$



|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $f'(x)$

1 point is earned for:  $f''(x)$

1 point is earned for:  $f'(-1) = 0$

1 point is earned for:  $f''(-2) = 0$

1 point is earned for:  $a$ ,  $b$

5-4

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$

**Part B**

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in  $k$

1 point is earned for  $k$

$$\begin{aligned} & \int_0^1 (4x^3 + 24x^2 + 36x + k) dx \\ &= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\ & 27 + k = 32 \\ & k = 5 \end{aligned}$$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

2 points are earned for antidifferentiation <-1> each error

1 point is earned for the expression in  $k$

1 point is earned for  $k$

**5-4**

$$\begin{aligned}\int_0^1 (4x^3 + 24x^2 + 36x + k) dx \\&= x^4 + 8x^3 + 18x^2 + kx \Big|_{x=0}^{x=1} = 27 + k \\27 + k &= 32 \\k &= 5\end{aligned}$$

## 5-4

28. A cubic polynomial function  $f$  is defined by

$$f(x) = 4x^3 + ax^2 + bx + k$$

where  $a$ ,  $b$ , and  $k$  are constants. The function  $f$  has a local minimum at  $x = -1$ , and the graph  $f$  has a point of inflection at  $x = -2$ .

Find the values of  $a$  and  $b$ .

**Part A**

1 point is earned for:  $f'(x)$

1 point is earned for:  $f''(x)$

1 point is earned for:  $f'(-1) = 0$

1 point is earned for:  $f''(-2) = 0$

1 point is earned for:  $a$ ,  $b$

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f'(-1) = 12 - 2a + b = 0$$

$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$



|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $f'(x)$

1 point is earned for:  $f''(x)$

1 point is earned for:  $f'(-1) = 0$

1 point is earned for:  $f''(-2) = 0$

1 point is earned for:  $a$ ,  $b$

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

**5-4**

$$f'(-1) = 12 - 2a + b = 0$$

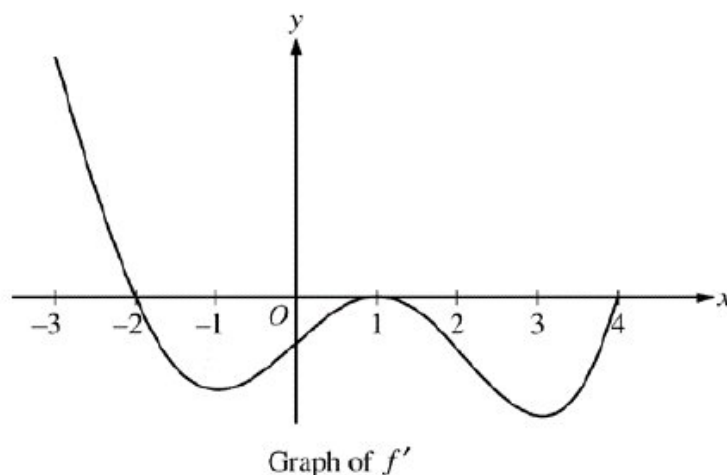
$$f''(-2) = -48 + 2a = 0$$

$$a = 24$$

$$b = -12 + 2a = 36$$

5-4

29.



The figure above shows the graph of  $f'$ , the derivative of a twice-differentiable function  $f$ , on the interval  $[-3, 4]$ . The graph of  $f'$  has horizontal tangents at  $x = -1$ ,  $x = 1$ , and  $x = 3$ . The areas of the regions bounded by the  $x$ -axis and the graph of  $f'$  on the intervals  $[-2, 1]$  and  $[1, 4]$  are 9 and 12, respectively.

- Find all  $x$ -coordinates at which  $f$  has a relative maximum. Give a reason for your answer.
- On what open intervals contained in  $-3 < x < 4$  is the graph of  $f$  both concave down and decreasing? Give a reason for your answer.
- Find the  $x$ -coordinates of all points of inflection for the graph of  $f$ . Give a reason for your answer.
- Given that  $f(1) = 3$ , write an expression for  $f(x)$  that involves an integral. Find  $f(4)$  and  $f(-2)$ .

### Part A

1 point is earned for identifying  $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$  at  $x = -2$ ,  $x = 1$ , and  $x = 4$ .

$f'(x)$  changes from positive to negative at  $x = -2$ .

Therefore,  $f$  has a relative maximum at  $x = -2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for identifying  $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$  at  $x = -2$ ,  $x = 1$ , and  $x = 4$ .

5-4

$f'(x)$  changes from positive to negative at  $x = -2$ .

Therefore,  $f$  has a relative maximum at  $x = -2$ .

**Part B**

1 point is earned for the intervals

1 point is earned for the reason

The graph of  $f$  is concave down and decreasing on the intervals  $-2 < x < -1$  and  $1 < x < 3$  because  $f'$  is decreasing and negative on these intervals.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The graph of  $f$  is concave down and decreasing on the intervals  $-2 < x < -1$  and  $1 < x < 3$  because  $f'$  is decreasing and negative on these intervals.

**Part C**

1 point is earned for identifying  $x = -1, 1$ , and  $3$

1 point is earned for the reason

The graph of  $f$  has a point of inflection at  $x = -1$  and  $x = 3$  because  $f'$  changes from decreasing to increasing at these points. The graph of  $f$  has a point of inflection at  $x = 1$  because  $f'$  changes from increasing to decreasing at this point.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for identifying  $x = -1, 1$ , and  $3$

1 point is earned for the reason

The graph of  $f$  has a point of inflection at  $x = -1$  and  $x = 3$  because  $f'$  changes from decreasing to increasing at these points. The graph of  $f$  has a point of inflection at  $x = 1$  because  $f'$  changes from increasing to decreasing at this point.

## 5-4

## Part D

1 point is earned for the integrand

1 point is earned for expressing  $f(x)$

1 point is earned for  $f(4)$  and  $f(-2)$

$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for expressing  $f(x)$

1 point is earned for  $f(4)$  and  $f(-2)$

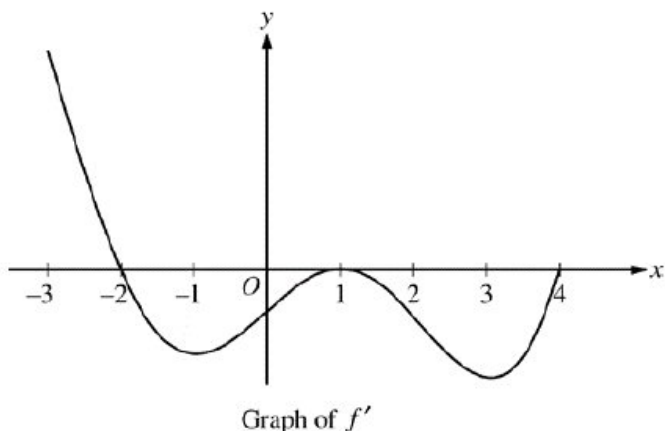
$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$

5-4

30.



The figure above shows the graph of  $f'$ , the derivative of a twice-differentiable function  $f$ , on the interval  $[-3, 4]$ . The graph of  $f'$  has horizontal tangents at  $x = -1$ ,  $x = 1$ , and  $x = 3$ . The areas of the regions bounded by the  $x$ -axis and the graph of  $f'$  on the intervals  $[-2, 1]$  and  $[1, 4]$  are 9 and 12, respectively.

Find all  $x$ -coordinates at which  $f$  has a relative maximum. Give a reason for your answer.

**Part A**

1 point is earned for identifying  $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$  at  $x = -2$ ,  $x = 1$ , and  $x = 4$ .  $f'(x)$  changes from positive to negative at  $x = -2$ .

Therefore,  $f$  has a relative maximum at  $x = -2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for identifying  $x = -2$

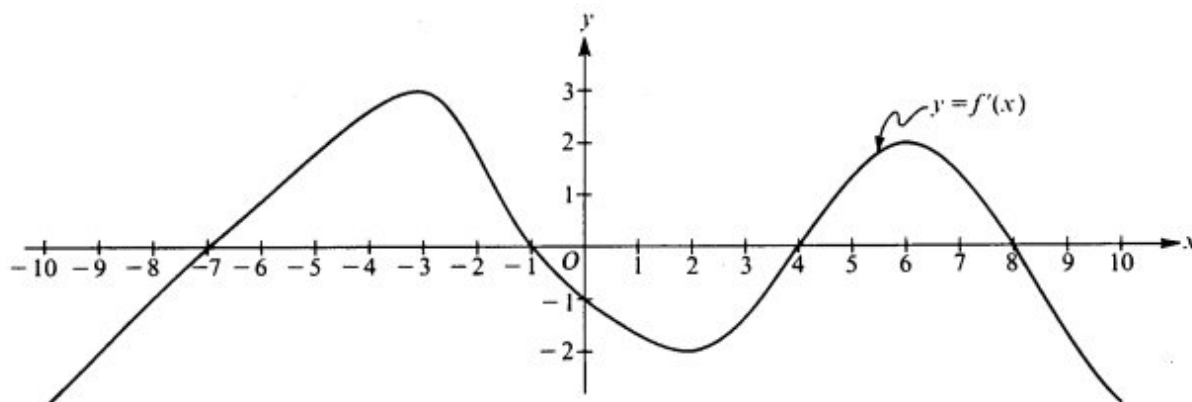
1 point is earned for answer with reason

$f'(x) = 0$  at  $x = -2$ ,  $x = 1$ , and  $x = 4$ .  $f'(x)$  changes from positive to negative at  $x = -2$ .

Therefore,  $f$  has a relative maximum at  $x = -2$ .

5-4

31.



Note: This is the graph of the derivative of  $f$ , not the graph of  $f$ .

The figure above shows the graph of  $f'$ , the derivative of a function  $f$ . The domain of  $f$  is the set of all real numbers  $x$  such that  $-10 \leq x \leq 10$ .

- For what values of  $x$  does the graph of  $f$  have a horizontal tangent?
- For what values of  $x$  in the interval  $(-10, 10)$  does  $f$  have a relative maximum? Justify your answer
- For what values of  $x$  is the graph of  $f$  concave downwards?

### Part A

3 point is earned for solving  $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent  $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

3 point is earned for solving  $f'(x) = 0$

1 point is **deducted** for each incorrect point

1 point is **deducted** for each missing point

horizontal tangent  $\Leftrightarrow f'(x) = 0$

$x = -7, -1, 4, 8$

5-4

**Part B**

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing pointRelative maxima at  $x = -1, 8$  because  $f'$  changes from positive to negative at these points

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing pointRelative maxima at  $x = -1, 8$  because  $f'$  changes from positive to negative at these points**Part C**

1 of 3 points are earned for correct sub intervals

2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

 $f$  concave downward  $\Rightarrow f'$  decreasing  $(-3, 2), (6, 10)$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all points:

1 of 3 points are earned for correct sub intervals

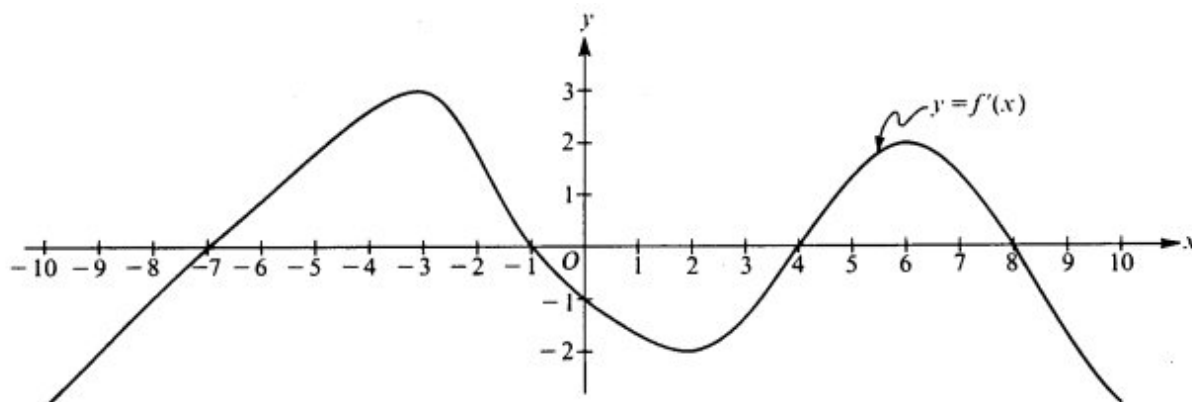
2 of 3 points are earned for 1 correct interval with analysis

3 of 3 points are earned for 2 correct intervals

 $f$  concave downward  $\Rightarrow f'$  decreasing  $(-3, 2), (6, 10)$

5-4

32.



Note: This is the graph of the derivative of  $f$ , not the graph of  $f$ .

The figure above shows the graph of  $f'$ , the derivative of a function  $f$ . The domain of  $f$  is the set of all real numbers  $x$  such that  $-10 \leq x \leq 10$ .

For what values of  $x$  in the interval  $(-10, 10)$  does  $f$  have a relative maximum? Justify your answer

### Part B

2 point is earned for answer

1 point is earned for justification

1 point is **deducted** for each incorrect or missing point

Relative maxima at  $x = -1, 8$  because  $f'$  changes from positive to negative at these points



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns three of the following points:

2 point is earned for answer

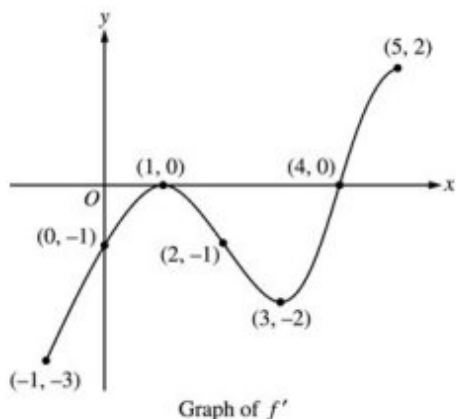
1 point is earned for justification

1 point is **deducted** for each incorrect or missing point

Relative maxima at  $x = -1, 8$  because  $f'$  changes from positive to negative at these points

5-4

33.



The figure above shows the graph of  $f'$ , the derivative of the function  $f$ , on the closed interval  $-1 \leq x \leq 5$ . The graph of  $f'$  has horizontal tangent lines at  $x=1$  and  $x=3$ . The function  $f$  is twice differentiable with  $f(2)=6$ .

At what value of  $x$  does  $f$  attain its absolute minimum value on the closed interval  $-1 \leq x \leq 5$ ? At what value of  $x$  does  $f$  attain its absolute maximum value on the closed interval  $-1 \leq x \leq 5$ ? Show the analysis that leads to your answers.

**Part B**

1 point is earned for indicating  $f$  decreases then increases

1 point is earned for eliminating  $x=5$  for maximum

1 point is earned for absolute minimum at  $x=4$

1 point is earned for absolute maximum at  $x=-1$

The function  $f$  decreases from  $x=-1$  to  $x=4$ , then increases from  $x=4$  to  $x=5$ .

Therefore, the absolute minimum value for  $f$  is at  $x=4$ . The absolute maximum value must occur at  $x=-1$  or at  $x=5$ .

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since  $f(5) < f(-1)$ , the absolute maximum value occurs at  $x=-1$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for indicating  $f$  decreases then increases

1 point is earned for eliminating  $x=5$  for maximum

1 point is earned for absolute minimum at  $x=4$

**5-4**

1 point is earned for absolute maximum at  $x=-1$

The function  $f$  decreases from  $x=-1$  to  $x=4$ , then increases from  $x=4$  to  $x=5$ .

Therefore, the absolute minimum value for  $f$  is at  $x=4$ . The absolute maximum value must occur at  $x=-1$  or at  $x=5$ .

$$f(5) - f(-1) = \int_{-1}^5 f(t) \underline{dt} < 0$$

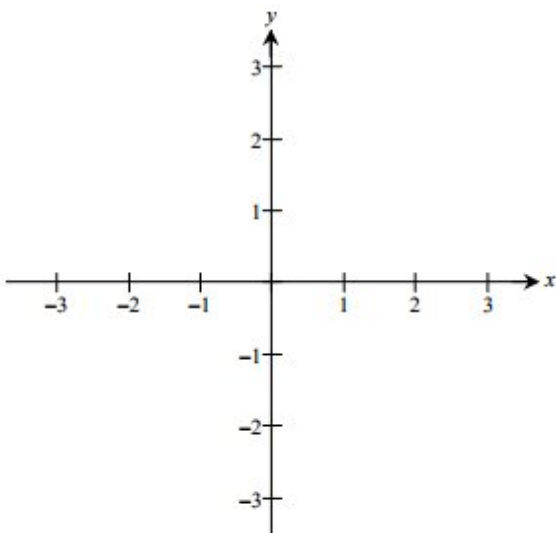
Since  $f(5) < f(-1)$ , the absolute maximum value occurs at  $x=-1$ .

5-4

34. Let  $f$  be a function that is even and continuous on the closed interval  $[-3, 3]$ . The function  $f$  and its derivatives have the properties indicated in the table below.

| $x$      | 0         | $0 < x < 1$ | 1 | $1 < x < 2$ | 2         | $2 < x < 3$ |
|----------|-----------|-------------|---|-------------|-----------|-------------|
| $f(x)$   | 1         | Positive    | 0 | Negative    | -1        | Negative    |
| $f'(x)$  | Undefined | Negative    | 0 | Negative    | Undefined | Positive    |
| $f''(x)$ | Undefined | Positive    | 0 | Negative    | Undefined | Negative    |

- (a) Find the  $x$ -coordinate of each point at which  $f$  attains an absolute maximum value or an absolute minimum value. For each  $x$ -coordinate you give, state whether  $f$  attains an absolute maximum or an absolute minimum.
- (b) Find the  $x$ -coordinate of each point of inflection on the graph of  $f$ . Justify your answer.
- (c) In the  $xy$ -plane provided below, sketch the graph of a function with all the given characteristics of  $f$ .

**Part A**

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at  $x = 0$

Absolute minimum at  $x = \pm 2$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

## 5-4

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at  $x = 0$

Absolute minimum at  $x = \pm 2$

**Part B**

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on  $[0,3)$  only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at  $x = \pm 1$  because the sign of  $f''(x)$  changes at  $x = 1$  and  $f$  is even.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on  $[0,3)$  only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at  $x = \pm 1$  because the sign of  $f''(x)$  changes at  $x = 1$  and  $f$  is even.

**Part C**

3 points are earned for sketch without error

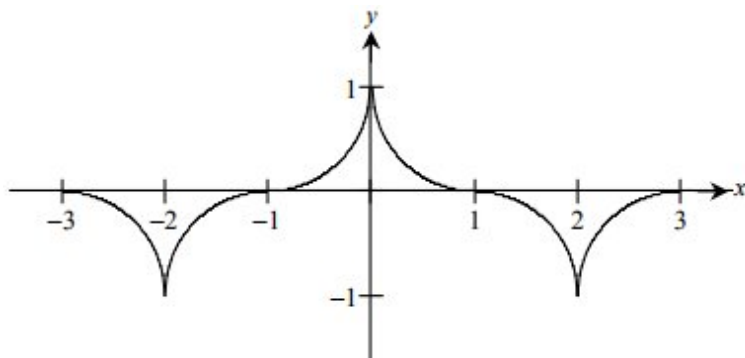
1 point is **deducted** for each error:

- symmetry
- slopes on  $[0,3)$

5-4

- concavity on  $[0,3)$
- discontinuity on  $[0,3)$
- $f(0), f(1), f(2)$  incorrect or extra intercept on  $[0,3)$

Note: No more than 1 point can be lost in any one category.



✓

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

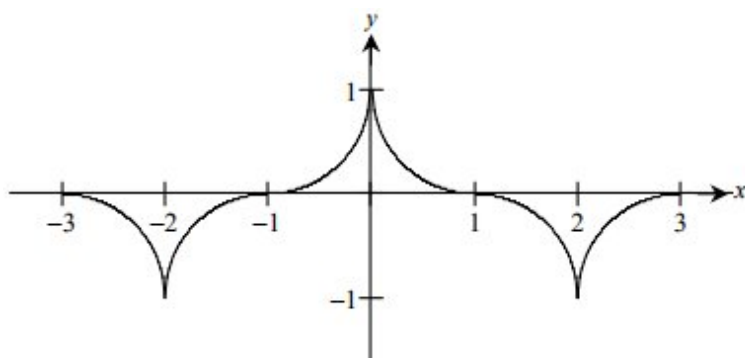
The student response earns all of the following points:

3 points are earned for sketch without error

1 point is **deducted** for each error:

- symmetry
- slopes on  $[0,3)$
- concavity on  $[0,3)$
- discontinuity on  $[0,3)$
- $f(0), f(1), f(2)$  incorrect or extra intercept on  $[0,3)$

Note: No more than 1 point can be lost in any one category.



5-4

5-4

35. Let  $f$  be the function given by  $f(x) = \frac{\ln x}{x}$  for all  $x > 0$ . The derivative of  $f$  is given by  $f'(x) = \frac{1 - \ln x}{x^2}$ .

- (a) Write an equation for the line tangent to the graph of  $f$  at  $x = e^2$ .
- (b) Find the  $x$ -coordinate of the critical point of  $f$ . Determine whether this point is a relative minimum, a relative maximum, or neither for the function  $f$ . Justify your answer.
- (c) The graph of the function  $f$  has exactly one point of inflection. Find the  $x$ -coordinate of this point.
- (d) Find  $\lim_{x \rightarrow 0^+} f(x)$ .

**Part A**

1 point is earned for the  $f(e^2)$  and  $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is  $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the  $f(e^2)$  and  $f'(e^2)$

1 point is earned for the answer

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is  $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$ .

## 5-4

## Part B

3 points can be earned.

1 point for  $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$  when  $x = e$ . The function  $f$  has a relative maximum at  $x = e$  because  $f'(x)$  changes from positive to negative at  $x = e$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

3 points can be earned.

1 point for  $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$  when  $x = e$ . The function  $f$  has a relative maximum at  $x = e$  because  $f'(x)$  changes from positive to negative at  $x = e$ .

## Part C

2 points are earned for  $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of  $f$  has a point of inflection at  $x = e^{3/2}$  because  $f''(x)$  changes sign at  $x = e^{3/2}$ .

5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for  $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

The graph of  $f$  has a point of inflection at  $x = e^{3/2}$  because  $f''(x)$  changes sign at  $x = e^{3/2}$ .

#### Part D

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$

5-4

36. Let  $f$  be the function given by  $f(x) = \frac{\ln x}{x}$  for all  $x > 0$ . The derivative of  $f$  is given by  $f'(x) = \frac{1 - \ln x}{x^2}$ . Find the  $x$ -coordinate of the critical point of  $f$ . Determine whether this point is a relative minimum, a relative maximum, or neither for the function  $f$ . Justify your answer.

**Part B**

3 points can be earned.

1 point for  $x = e$

1 point for relative maximum

1 point for justification.

$f'(x) = 0$  when  $x = e$ . The function  $f$  has a relative maximum at  $x = e$  because  $f'(x)$  changes from positive to negative at  $x = e$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns three of the following points:

3 points can be earned.

1 point for  $x = e$

1 point for relative maximum

1 point for justification.

$f'(x) = 0$  when  $x = e$ . The function  $f$  has a relative maximum at  $x = e$  because  $f'(x)$  changes from positive to negative at  $x = e$ .

## 5-4

37. Let  $f$  be the function given by  $f(x) = x^3 - 6x^2 + p$ , where  $p$  is an arbitrary constant.
- (a) Write an expression for  $f'(x)$  and use it to find the relative maximum and minimum values of  $f$  in terms of  $p$ . Show the analysis that leads to your conclusion.
- (b) For what values of the constant  $p$  does  $f$  have 3 distinct roots?
- (c) Find the value of  $p$  such that the average value of  $f$  over the closed interval  $[-1, 2]$  is 1.

**Part A**

1 point is earned for correctly finding  $f'(x)$

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving  $f'(x) = 0$

$$3x(x - 4) = 0$$

$$x = 0, x = 4$$

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for  $p$

$f'(x)$  changes sign from positive to negative at  $x = 0$

$f'(x)$  changes sign from negative to positive at  $x = 4$

or

$$f''(x) = 6x - 12, f''(0) = -12, f''(4) = 12$$

1 point is earned for correctly finding the maximum and minimum values in terms of  $p$

relative maximum at  $x = 0, f(0) = p$

relative minimum at  $x = 4, f(4) = p - 32$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for correctly finding  $f'(x)$

## 5-4

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving  $f'(x) = 0$

$$3x(x - 4) = 0$$

$$x = 0, x = 4$$

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for  $p$

$f(x)$  changes sign from positive to negative at  $x = 0$

$f(x)$  changes sign from negative to positive at  $x = 4$

or

$$f''(x) = 6x - 12, f''(0) = -12, f''(4) = 12$$

1 point is earned for correctly finding the maximum and minimum values in terms of  $p$

relative maximum at  $x = 0, f(0) = p$

relative minimum at  $x = 4, f(4) = p - 32$

**Part B**

1 point is earned for the correct upper bound

1 point is earned for the correct lower bound

$<-1>$  including 0 or 32, or both

$f(x)$  has three distinct real roots when  $p > 0$  and  $p - 32 < 0$ ,

so  $0 < p < 32$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct upper bound

1 point is earned for the correct lower bound

## 5-4

$<-1>$  including 0 or 32, or both

$f(x)$  has three distinct real roots when  $p > 0$  and  $p - 32 < 0$ ,

so  $0 < p < 32$

**Part C**

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for  $p$  where the equation must be of the

form  $k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$

$$\frac{1}{2 - (-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[ \frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[ \left( \frac{16}{4} - 16 + 2p \right) - \left( \frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[ 3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for the average value where 1 point is earned for the correct integrand and limits and 1 point is earned for the appropriate constant for definite integral

1 point is earned for correctly setting average value equal to 1 and solving for  $p$  where the equation must be of the

form  $k \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$

**5-4**

$$\frac{1}{2-(-1)} \int_{-1}^2 (x^3 - 6x^2 + p) dx = 1$$

$$\frac{1}{3} \left[ \frac{1}{4} x^4 - 2x^3 + px \right]_{-1}^2 = 1$$

$$\frac{1}{3} \left[ \left( \frac{16}{4} - 16 + 2p \right) - \left( \frac{1}{4} + 2 - p \right) \right]$$

$$= \frac{1}{3} \left[ 3p - \frac{57}{4} \right] = 1$$

$$p = \frac{23}{4} = 5.75$$

1/3 solution only

## 5-4

38. Let  $f$  be the function given by  $f(x) = x^3 - 6x^2 + p$ , where  $p$  is an arbitrary constant.

Write an expression for  $f'(x)$  and use it to find the relative maximum and minimum values of  $f$  in terms of  $p$ . Show the analysis that leads to your conclusion.

## Part A

1 point is earned for correctly finding  $f'(x)$

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving  $f'(x) = 0$

$$3x(x-4) = 0$$

$$x=0, x=4$$

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for  $p$

$f'(x)$  changes sign from positive to negative at  $x=0$

$f'(x)$  changes sign from negative to positive at  $x=4$

or

$$f''(x) = 6x - 12, f''(0) = -12, f''(4) = 12$$

1 point is earned for correctly finding the maximum and minimum values in terms of  $p$

relative maximum at  $x=0, f(0) = p$

relative minimum at  $x=4, f(4) = p - 32$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns four of the following points:

1 point is earned for correctly finding  $f'(x)$

$$f(x) = x^3 - 6x^2 + p$$

$$f'(x) = 3x^2 - 12x = 0$$

1 point is earned for correctly solving  $f'(x) = 0$

$$3x(x-4) = 0 \quad x=0, x=4$$

**5-4**

1 point is earned for correctly indicating the location of a maximum and a minimum, with analysis

0/1 if explicitly chooses to work only with a specific value for  $p$

$f'(x)$  changes sign from positive to negative at  $x=0$

$f'(x)$  changes sign from negative to positive at  $x=4$

or

$f''(x)=6x-12, f''(0)=-12, f''(4)=12$

1 point is earned for correctly finding the maximum and minimum values in terms of  $p$

relative maximum at  $x=0, f(0)=p$

relative minimum at  $x=4, f(4)=p-32$

5-4

39. Let  $f$  be the function given by  $f(x) = \frac{2x}{\sqrt{x^2+x+1}}$ .

Find the range of  $f$ . Use  $f'(x)$  to justify your answer.

Note:  $f'(x) = x+2 / (x^2+x+1)^{3/2}$

**Part D**

1 point is earned for the correct right endpoint

1 point is earned for the correct left endpoint (min)

1 point is earned for the correct inclusion/exclusion of endpoints

Range is  $\left[\frac{-4}{\sqrt{3}}, 2\right)$  or  $-2.309 \leq y < 2$

1 point is earned for using sign of  $f'$  to justify

Sign of  $f'$  is given by  $\frac{-}{-2} \frac{+}{+}$ , implying that  $f$  has a minimum at  $x = -2$  and, with asymptotes, that  $f$  is never greater than 2.

Note: max 2/4 (1 - 1 - 0 - 0) if using incorrect asymptotes



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns four of the following points:

1 point is earned for the correct right endpoint

1 point is earned for the correct left endpoint (min)

1 point is earned for the correct inclusion/exclusion of endpoints

Range is  $\left[\frac{-4}{\sqrt{3}}, 2\right)$  or  $-2.309 \leq y < 2$

1 point is earned for using sign of  $f'$  to justify

5-4

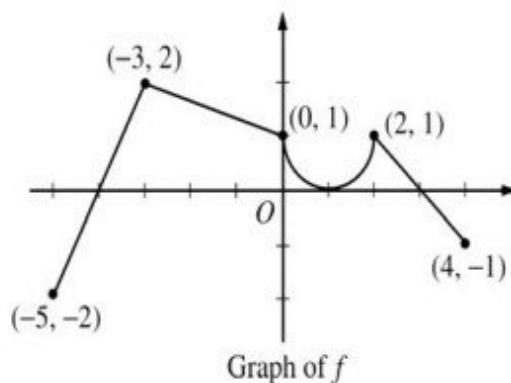
Sign of  $f'$  is given by  $\begin{array}{c} - \qquad \qquad + \\ \hline -2 \end{array}$ , implying that  $f$  has a minimum at  $x = -2$  and, with

asymptotes, that  $f$  is never greater than 2.

Note: max 2/4 (1 - 1 - 0 - 0) if using incorrect asymptotes

5-4

40.



The graph of the function  $f$  shown above consists of a semicircle and three line segments. Let  $g$  be the function given by  $g(x) = \int_{-3}^x f(t) dt$ .

- Find  $g(0)$  and  $g'(0)$ .
- Find all values of  $x$  in the open interval  $(-5, 4)$  at which  $g$  attains a relative maximum. Justify your answer.
- Find the absolute minimum value of  $g$  on the closed interval  $[-5, 4]$ . Justify your answer.
- Find all values of  $x$  in the open interval  $(-5, 4)$  at which the graph of  $g$  has a point of inflection.

**Part A**

1 point is earned for  $g(0)$

1 point is earned for  $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $g(0)$

1 point is earned for  $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$

## 5-4

## Part B

1 point is earned for  $x = 3$

1 point is earned for justification

$g$  has a relative maximum at  $x = 3$ .

This is the only  $x$ -value where  $g' = f$  changes from positive to negative.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $x = 3$

1 point is earned for justification

$g$  has a relative maximum at  $x = 3$ .

This is the only  $x$ -value where  $g' = f$  changes from positive to negative.

## Part C

1 point is earned for identifying  $x = -4$  as a candidate

1 point is earned for  $g(-4) = -1$

1 point is earned for justification and answer

The only  $x$ -value where  $f$  changes from negative to positive is  $x = -4$ . The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of  $g$  is  $-1$ .

5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for identifying  $x = -4$  as a candidate

1 point is earned for  $g(-4) = -1$

1 point is earned for justification and answer

The only  $x$ -value where  $f$  changes from negative to positive is  $x = -4$ . The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of  $g$  is  $-1$ .

#### Part D

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

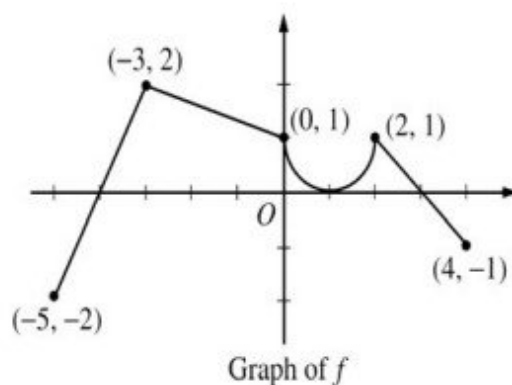
Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$

5-4

41.



The graph of the function  $f$  shown above consists of a semicircle and three line segments. Let  $g$  be the function given by  $g(x) = \int_{-3}^x f(t) dt$ .

Find all values of  $x$  in the open interval  $(-5, 4)$  at which  $g$  attains a relative maximum. Justify your answer.

**Part B**

1 point is earned for  $x = 3$

1 point is earned for justification

$g$  has a relative maximum at  $x = 3$ .

This is the only  $x$ -value where  $g' = f$  changes from positive to negative.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $x = 3$

1 point is earned for justification

$g$  has a relative maximum at  $x = 3$ .

This is the only  $x$ -value where  $g' = f$  changes from positive to negative.

## 5-4

42. Consider a differentiable function  $f$  having domain all positive real numbers, and for which it is known that  $f'(x) = (4-x)x^{-3}$  for  $x > 0$ .

Find the  $x$ -coordinate of the critical point of  $f$ . Determine whether the point is a relative maximum, a relative minimum, or neither for the function  $f$ . Justify your answer.

## Part A

1 point is earned for  $x=4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x)=0 \text{ at } x=4$$

$$f'(x)>0 \text{ for } 0<x<4$$

$$f'(x)<0 \text{ for } x>4$$

Therefore  $f$  has a relative maximum at  $x=4$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $x=4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x)=0 \text{ at } x=4$$

$$f'(x)>0 \text{ for } 0<x<4$$

$$f'(x)<0 \text{ for } x>4$$

Therefore  $f$  has a relative maximum at  $x=4$ .

## 5-4

43. Consider a differentiable function  $f$  having domain all positive real numbers, and for which it is known that  $f'(x) = (4 - x)x^{-3}$  for  $x > 0$ .
- (a) Find the  $x$ -coordinate of the critical point of  $f$ . Determine whether the point is a relative maximum, a relative minimum, or neither for the function  $f$ . Justify your answer.
- (b) Find all intervals on which the graph of  $f$  is concave down. Justify your answer.
- (c) Given that  $f(1) = 2$ , determine the function  $f$ .

## Part A

1 point is earned for  $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore  $f$  has a relative maximum at  $x = 4$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $x = 4$

1 point is earned for relative maximum

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = 4$$

$$f'(x) > 0 \text{ for } 0 < x < 4$$

$$f'(x) < 0 \text{ for } x > 4$$

Therefore  $f$  has a relative maximum at  $x = 4$ .

## Part B

2 points are earned for  $f''(x)$

1 point is earned for answer with justification

5-4

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of  $f$  is concave down on the interval  $0 < x < 6$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for  $f''(x)$

1 point is earned for answer with justification

$$\begin{aligned}
 f''(x) &= -x^{-3} + (4 - x)(-3x^{-4}) \\
 &= -x^{-3} - 12x^{-4} + 3x^{-3} \\
 &= 2x^{-4}(x - 6) \\
 &= \frac{2(x-6)}{x^4}
 \end{aligned}$$

$$f''(x) < 0 \text{ for } 0 < x < 6$$

The graph of  $f$  is concave down on the interval  $0 < x < 6$ .

### Part C

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for the answer

$$\begin{aligned}
 f(x) &= 2 + \int_1^x (4t^{-3} - t^{-2}) dt \\
 &= 2 + [-2t^{-2} + t^{-1}]_{t=1}^{t=x} \\
 &= 3 - 2x^{-2} + x^{-1}
 \end{aligned}$$

5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for antiderivative

1 point is earned for the answer

$$\begin{aligned}f(x) &= 2 + \int_1^x (4t^{-3} - t^{-2}) dt \\&= 2 + [-2t^{-2} + t^{-1}]_{t=1}^{t=x} \\&= 3 - 2x^{-2} + x^{-1}\end{aligned}$$

## 5-4

44. Let  $f$  be the function given by  $f(x) = x^3 - 5x^2 + 3x + k$ , where  $k$  is a constant.

(a) On what intervals is  $f$  increasing?

(b) On what intervals is the graph of  $f$  concave downward?

(c) Find the value of  $k$  for which  $f$  has 11 as its relative minimum.

**Part A**

1 point is earned for the correct  $f'(x)$

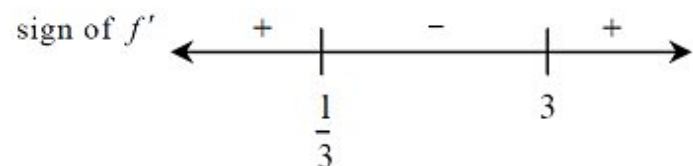
$$\begin{aligned} f'(x) &= 3x^2 - 10x + 3 \\ &= (3x - 1)(x - 3) \end{aligned}$$

1 point is earned for the correct zeros of  $f'(x)$

$$f'(x) = 0 \text{ at } x = 1/3 \text{ and } x = 3$$

1 point is earned for correctly analyzing the sign of  $f'$

Note: 0/1 if  $f'$  has fewer than two zeros or if the zeros are incorrect for  $f'$



1 point is earned for correctly indicating answer is where  $f'(x) > 0$

$f$  is increasing on  $(-\infty, 1/3]$  and on  $[3, \infty)$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct  $f'(x)$

$$\begin{aligned} f'(x) &= 3x^2 - 10x + 3 \\ &= (3x - 1)(x - 3) \end{aligned}$$

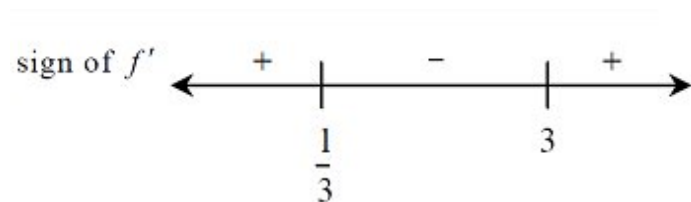
1 point is earned for the correct zeros of  $f'(x)$

$$f'(x) = 0 \text{ at } x = 1/3 \text{ and } x = 3$$

## 5-4

1 point is earned for correctly analyzing the sign of  $f'$

Note: 0/1 if  $f'$  has fewer than two zeros or if the zeros are incorrect for  $f'$



1 point is earned for correctly indicating answer is where  $f'(x) > 0$

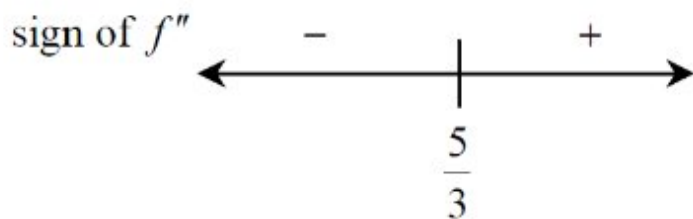
$f$  is increasing on  $(-\infty, 1/3]$  and on  $[3, \infty)$

**Part B**

1 point is earned for the correct  $f''(x)$

$$f''(x) = 6x - 10$$

1 point is earned for correctly analyzing sign of  $f''$  OR sets  $f''(x) < 0$



1 point is earned for correctly choosing interval where  $f''(x) < 0$

graph is concave down on  $(-\infty, \frac{5}{3})$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

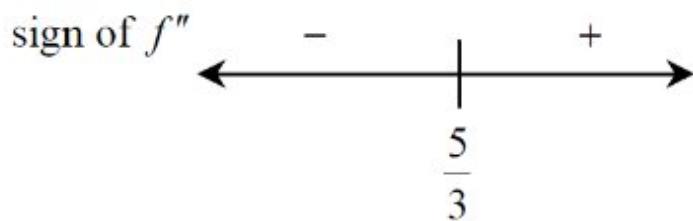
The student response earns all of the following points:

1 point is earned for the correct  $f''(x)$

$$f''(x) = 6x - 10$$

1 point is earned for correctly analyzing sign of  $f''$  OR sets  $f''(x) < 0$

5-4



1 point is earned for correctly choosing interval where  $f''(x) < 0$

graph is concave down on  $(-\infty, \frac{5}{3})$

**Part C**

1 point is earned for the correct evaluation of the relative minimum

from (a),  $f$  has its relative minimum at  $x = 3$

1 point is earned for correctly setting relative min. = 11 and solves for  $k$

so,

$$f(3) = (3)^3 - 5(3)^2 + 3(3) + k$$

$$= -9 + k = 11$$

$$k = 20$$

Note: 0/2 if evaluates  $f$  at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the correct evaluation of the relative minimum

from (a),  $f$  has its relative minimum at  $x = 3$

1 point is earned for correctly setting relative min. = 11 and solves for  $k$

so,

**5-4**

$$f(3) = (3)^3 - 5(3)^2 + 3(3) + k$$

$$= -9 + k = 11$$

$$k = 20$$

Note: 0/2 if evaluates  $f$  at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves

## 5-4

45. Let  $f$  be the function given by  $f(x) = x^3 - 5x^2 + 3x + k$ , where  $k$  is a constant. Find the value of  $k$  for which  $f$  has 11 as its relative minimum.

## Part C

1 point is earned for the correct evaluation of the relative minimum

from (a),  $f$  has its relative minimum at  $x=3$

1 point is earned for correctly setting relative min. = 11 and solves for  $k$

so,

$$f(3) = (3)^3 - 5(3)^2 + 3(3) + k$$

$$= -9 + k = 11$$

$$k = 20$$

Note: 0/2 if evaluates  $f$  at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for the correct evaluation of the relative minimum

from (a),  $f$  has its relative minimum at  $x=3$

1 point is earned for correctly setting relative min. = 11 and solves for  $k$

so,

$$f(3) = (3)^3 - 5(3)^2 + 3(3) + k$$

Note: 0/2 if evaluates  $f$  at a non-critical number

1/2 if student's work shows no relative min. and student so states

1/2 if evaluates at relative max and correctly solves

5-4

46. A particle moves along the x-axis so that its acceleration at any time  $t$  is  $a(t) = 2t - 7$ . If the initial velocity of the particle is 6, at what time  $t$  during the interval  $0 \leq t \leq 4$  is the particle farthest to the right?

(A) 0

(B) 1



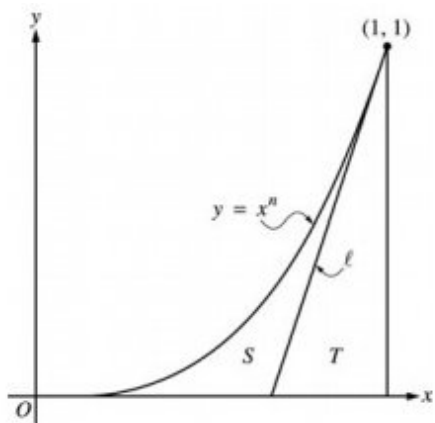
(C) 2

(D) 3

(E) 4

5-4

47.



Let  $l$  be the line tangent to the graph of  $y = x^n$  at the point  $(1, 1)$ , where  $n > 1$ , as shown above.

Let  $S$  be the region bounded by the graph of  $y = x^n$ , the line  $l$ , and the  $x$ -axis. Express the area of  $S$  in terms of  $n$  and determine the value of  $n$  that maximizes the area of  $S$ .

**Part C**

1 point is earned for the area of  $S$  in terms of  $n$

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for  $n$

$$\begin{aligned}
 \text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\
 &= \frac{1}{n+1} - \frac{1}{2n} \\
 \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\
 2n^2 &= (n+1)^2 \\
 \sqrt{2}n &= (n+1) \\
 n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}
 \end{aligned}$$

✓

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for the area of  $S$  in terms of  $n$

1 point is earned for the derivative

## 5-4

1 point is earned if sets derivative equal to 0

1 point is earned if solves for  $n$

$$\begin{aligned} \text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2} \end{aligned}$$

48. Consider the differential equation  $\frac{dy}{dx} = 2x - y$

Let  $y=f(x)$  be the particular solution to the differential equation with the initial condition  $f(2)=3$ . Does  $f$  have a relative minimum, a relative maximum, or neither at  $x=2$ ? Justify your answer.

## Part C

1 point is earned for considering  $\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore,  $f$  has neither a relative minimum nor a relative maximum at  $x=2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for considering  $\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

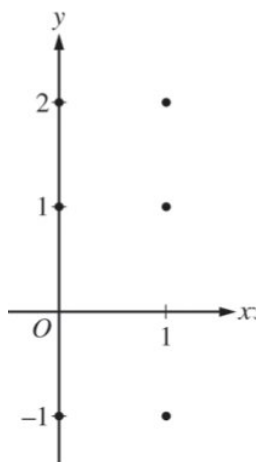
$$\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore,  $f$  has neither a relative minimum nor a relative maximum at  $x=2$ .

5-4

49. Consider the differential equation  $\frac{dy}{dx} = 2x - y$ .

(a) On the axes provided, sketch a slope field for the given differential equation at the six points indicated.



(b) Find  $\frac{d^2y}{dx^2}$  in terms of  $x$  and  $y$ . Determine the concavity of all solution curves for the given differential equation in Quadrant II. Give a reason for your answer.

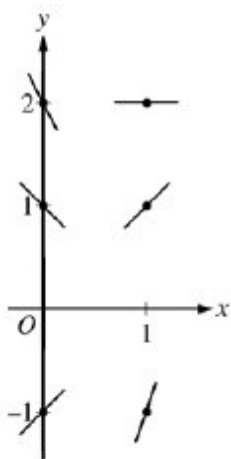
(c) Let  $y = f(x)$  be the particular solution to the differential equation with the initial condition  $f(2) = 3$ . Does  $f$  have a relative minimum, a relative maximum, or neither at  $x = 2$ ? Justify your answer.

(d) Find the values of the constants  $m$  and  $b$  for which  $y = mx + b$  is a solution to the differential equation.

### Part A

1 point is earned for slopes where  $x = 0$

1 point is earned for slopes where  $x = 1$



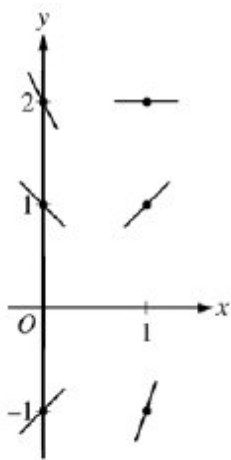
|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

5-4

1 point is earned for slopes where  $x = 0$

1 point is earned for slopes where  $x = 1$

**Part B**

1 point is earned for  $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II,  $x < 0$  and  $y > 0$ , so  $2 - 2x + y > 0$ . Therefore, all solution curves are concave up in Quadrant II.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II,  $x < 0$  and  $y > 0$ , so  $2 - 2x + y > 0$ . Therefore, all solution curves are concave up in Quadrant II.

**Part C**

## 5-4

1 point is earned for considering  $\frac{dy}{dx}\big|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\frac{dy}{dx}\big|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore,  $f$  has neither a relative minimum nor a relative maximum at  $x = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for considering  $\frac{dy}{dx}\big|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\frac{dy}{dx}\big|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore,  $f$  has neither a relative minimum nor a relative maximum at  $x = 2$ .

**Part D**

1 point is earned for  $\frac{d}{dx}(mx + b) = m$

1 point is earned for  $2x - y = m$

1 point is earned for answer

$$y = mx + b \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(mx + b) = m$$

$$2x - y = m$$

$$2x - (mx + b) = m$$

$$(2 - m)x - (m + b) = 0$$

$$2 - m = 0 \Rightarrow m = 2$$

$$b = -m \Rightarrow b = -2$$

Therefore,  $m = 2$ , and  $b = -2$

5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $\frac{d}{dx}(mx + b) = m$

1 point is earned for  $2x - y = m$

1 point is earned for answer

$$y = mx + b \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(mx + b) = m$$

$$2x - y = m$$

$$2x - (mx + b) = m$$

$$(2 - m)x - (m + b) = 0$$

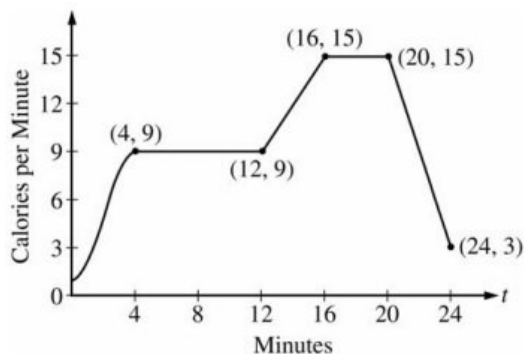
$$2 - m = 0 \Rightarrow m = 2$$

$$b = -m \Rightarrow b = -2$$

Therefore,  $m = 2$ , and  $b = -2$

5-4

50.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function  $f$ . In the figure above,  $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$  for  $0 \leq t \leq 4$  and  $f$  is piecewise linear for  $4 \leq t \leq 24$ .

- Find  $f'(22)$ . Indicate units of measure.
- For the time interval  $0 \leq t \leq 24$ , at what time  $t$  is  $f$  increasing at its greatest rate? Show the reasoning that supports your answer.
- Find the total number of calories burned over the time interval  $6 \leq t \leq 18$  minutes.
- The setting on the machine is now changed so that the person burns  $f(t) + c$  calories per minute. For this setting, find  $c$  so that an average of 15 calories per minute is burned during the time interval  $6 \leq t \leq 18$ .

**Part A**

1 point is earned for  $f'(22)$  and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$



0

1

The student response earns all of the following points:

1 point is earned for  $f'(22)$  and units

$$f'(22) = \frac{15 - 3}{20 - 24} = -3 \text{ calories/min/min}$$

**Part B**

1 point is earned for  $f'$  on  $(0, 4)$

## 5-4

1 point is earned for showing  $f'$  has a max at  $t = 2$  on  $(0, 4)$

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12, 16)$ ,  $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0, 4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f'$  has a maximum on  $[0, 4]$  since  $f'' > 0$  on  $(0, 2)$  and  $f'' < 0$  on  $(2, 4)$ .

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $f'$  on  $(0, 4)$

1 point is earned for showing  $f'$  has a max at  $t = 2$  on  $(0, 4)$

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12, 16)$ ,  $f'(t) = \frac{15 - 9}{16 - 12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0, 4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f'$  has a maximum on  $[0, 4]$  since  $f'' > 0$  on  $(0, 2)$  and  $f'' < 0$  on  $(2, 4)$ .

5-4

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$ .

**Part C**

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

$$\int_6^{18} f(t) dt = 6(9) + \frac{1}{2}(4)(9 + 15) + 2(15) = 132 \text{ calories}$$

**Part D**

1 point is earned for the setup

1 point is earned for the value of  $c$

$$\text{We want } \frac{1}{12} \int_6^{18} (f(t) + c) dt = 15.$$

This means  $132 + 12c = 15(12)$ . So,  $c = 4$ .

OR

Currently, the average is  $\frac{132}{12} = 11$  calories/min.

Adding  $c$  to  $f(t)$  will shift the average by  $c$ . So  $c = 4$  to get an average of 15 calories/min.

5-4



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns none of the following points:

1 point is earned for the setup

1 point is earned for the value of  $c$

We want  $\frac{1}{12} \int_6^{18} (f(t) + c) dt = 15$ .

This means  $132 + 12c = 15(12)$ . So,  $c = 4$ .

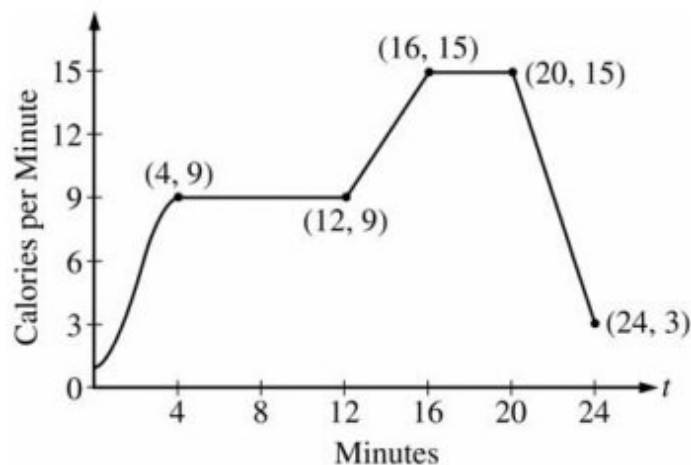
OR

Currently, the average is  $\frac{132}{12} = 11$  calories/min.

Adding  $c$  to  $f(t)$  will shift the average by  $c$ . So  $c = 4$  to get an average of 15 calories/min.

5-4

51.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function  $f$ . In the figure above,  $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$  for  $0 \leq t \leq 4$  and  $f$  is piecewise linear for  $4 \leq t \leq 24$ .

For the time interval  $0 \leq t \leq 24$ , at what time  $t$  is  $f$  increasing at its greatest rate? Show the reasoning that supports your answer.

**Part B**

1 point is earned for  $f'$  on  $(0,4)$

1 point is earned for showing  $f'$  has a max at  $t = 2$  on  $(0,4)$

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12,16)$ ,  $f'(t) = \frac{15-9}{16-12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0,4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f'$  has a maximum on  $[0, 4]$

since  $f'' > 0$  on  $(0,2)$  and  $f'' < 0$  on  $(2, 4)$ .

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $f'$  on  $(0,4)$

## 5-4

1 point is earned for showing  $f'$  has a max at  $t = 2$  on  $(0, 4)$

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12, 16)$ ,  $f'(t) = \frac{15-9}{16-12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0, 4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f'$  has a maximum on  $[0, 4]$

since  $f'' > 0$  on  $(0, 2)$  and  $f'' < 0$  on  $(2, 4)$ .

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$

52. The function  $f$  given by  $f(x) = 3x^5 - 4x^3 - 3x$  has a relative maximum at  $x =$

(A) -1

(B)  $-\frac{\sqrt{5}}{5}$

(C) 0

(D)  $\frac{\sqrt{5}}{5}$

(E) 1



53.  If  $f'(x) = \sqrt{x^4 + 1} + x^3 - 3x$ , then  $f$  has a local maximum at  $x =$

(A) -2.314

(B) -1.332

(C) 0.350

(D) 0.829

(E) 1.234



54.  Let  $f$  be the function with derivative given by  $f'(x) = \sin(x^2 + 1)$ . How many relative extrema does  $f$  have on the interval  $2 < x < 4$ ?

(A) One

(B) Two

(C) Three

(D) Four

(E) Five



## 5-4

55. The number of gallons,  $P(t)$ , of a pollutant in a lake changes at the rate  $P'(t) = 1 - 3e^{-0.2\sqrt{t}}$  gallons per day, where  $t$  is measured in days. There are 50 gallons of the pollutant in the lake at time  $t = 0$ . The lake is considered to be safe when it contains 40 gallons or less of pollutant.



For what value of  $t$  will the number of gallons of pollutant be at its minimum? Justify your answer.

**Part B**

1 point is earned for sets  $P'(t) = 0$

1 point is earned for solves for  $t$

1 point is earned for the justification

$$P'(t) = 1 - 3e^{-0.2\sqrt{t}} = 0$$

$$t = (5 \ln 3)^2 = 30.174$$

$P'(t)$  is negative for  $0 < t < (5 \ln 3)^2$  and positive for  $t > (5 \ln 3)^2$ . There for there is a minimum at  $t = (5 \ln 3)^2$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for sets  $P'(t) = 0$

1 point is earned for solves for  $t$

1 point is earned for the justification

$$P'(t) = 1 - 3e^{-0.2\sqrt{t}} = 0$$

$$t = (5 \ln 3)^2 = 30.174$$

$P'(t)$  is negative for  $0 < t < (5 \ln 3)^2$  and positive for  $t > (5 \ln 3)^2$ . There for there is a minimum at  $t = (5 \ln 3)^2$ .

## 5-4

56. This problem deals with functions defined by  $f(x) = x + b \sin x$ , where  $b$  is a positive constant and  $-2\pi \leq x \leq 2\pi$ .

Are the points of tangency described in part (b) relative maximum points of  $f$ ? Why?

## Part C

1 point is earned for the correct answer with reason

No, because  $f'(x) = 1$  (or  $f'(x) \neq 0$ ) at  $x$ -coordinates of points of tangency



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns one of the following points:

1 point is earned for the correct answer with reason

No, because  $f'(x) = 1$  (or  $f'(x) \neq 0$ ) at  $x$ -coordinates of points of tangency

57. For what value of  $k$  will  $x + k/x$  have a relative maximum at  $x = -2$ ?

- (A) -4  
(B) -2  
(C) 2

(D) 4



(E) None of these

58. A point moves on the  $x$ -axis in such a way that its velocity at time  $t$  ( $t > 0$ ) is given by  $v = \ln t/t$ . At what value of  $t$  does  $v$  attain its maximum?

- (A) 1  
(B)  $e^{1/2}$

(C)  $e$



(D)  $e^{3/2}$

(E) There is no maximum value for  $v$ .

59. The derivative of  $f(x) = \frac{x^4}{3} - \frac{x^5}{5}$  attains its maximum value at  $x =$

- (A) -1  
(B) 0

(C) 1



(D)  $\frac{4}{3}$

(E)  $\frac{5}{3}$

5-4

60. The point on the curve  $2y=x^2$  nearest to  $(4,1)$  is

(A)  $(0,0)$

(B)  $(2,2)$  ✓

(C)  $(\sqrt{2}, 1)$

(D)  $(2\sqrt{2}, 4)$

(E)  $(4,8)$

61. The function defined by  $f(x) = x^3 - 3x^2$  for all real numbers  $x$  has a relative maximum at  $x =$

(A)  $-2$

(B)  $0$  ✓

(C)  $1$

(D)  $2$

(E)  $4$

62. At what values of  $x$  does  $f(x) = 3x^5 - 5x^3 + 15$  have a relative maximum?

(A)  $-1$  only ✓

(B)  $0$  only

(C)  $1$  only

(D)  $-1$  and  $1$  only

(E)  $-1, 0$  and  $1$

63. For what value of  $x$  does the function  $f(x) = (x - 2)(x - 3)^2$  have a relative maximum?

(A)  $-3$

(B)  $-7/3$

(C)  $-5/2$

(D)  $7/3$  ✓

(E)  $5/2$

64. What is the minimum value of  $f(x) = x \ln(x)$ ?

(A)  $-e$

(B)  $-1$

(C)  $-\frac{1}{e}$  ✓

(D)  $0$

(E) has no minimum value

5-4

65. A particle moves along the  $x$ -axis so that at any time  $t > 0$  its velocity is given by  $v(t) = t \ln t - t$ . At time  $t = 1$ , the position of the particle is  $x(1) = 6$ .

What is the minimum velocity of the particle? Show the analysis that leads to your conclusion.

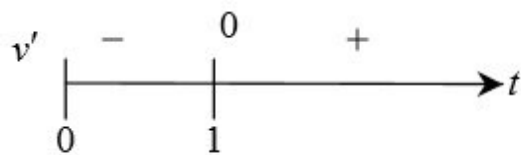
**Part C**

1 point is earned for correctly solving  $v'(t) = 0$

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is  $v(1) = -1$

Max 1/3 if student's  $v'$  does not involve  $\ln$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

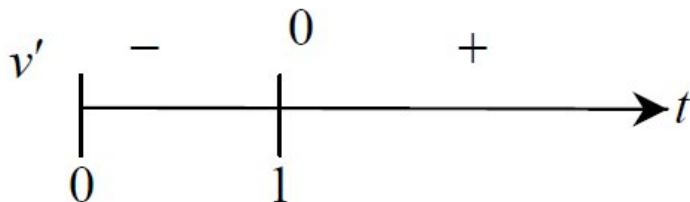
The student response earns three of the following points:

1 point is earned for correctly solving  $v'(t) = 0$

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

5-4

minimum velocity is  $v(1) = -1$

Max 1/3 if student's  $v'$  does not involve  $\ln$

66.  The derivative of  $f$  is  $x^4(x - 2)(x + 3)$ . At how many points will the graph of  $f$  have a relative maximum?

(A) None

(B) One

(C) Two

(D) Three

(E) Four



5-4

67. A particle moves along the  $y$ -axis so that its velocity  $v$  at time  $t \geq 0$  is given by  $v(t) = 1 - \tan^{-1}(e^t)$ . At time  $t = 0$ , the particle is at  $y = -1$ . (Note:  $\tan^{-1}x = \arctan x$ )



Find the time  $t \geq 0$  at which the particle reaches its highest point. Justify your answer.

**Part C**

1 point is earned for: set  $v(t) = 0$

1 point is earned for identifies  $t = 0.433$  as a candidate

1 point is earned for justifies absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$  is the only critical value for  $y$ .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$  has an absolute maximum at  $t = 0.433$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for: set  $v(t) = 0$

1 point is earned for identifies  $t = 0.433$  as a candidate

1 point is earned for justifies absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$  is the only critical value for  $y$ .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$  has an absolute maximum at  $t = 0.433$ .

## 5-4

68. Mighty Cable Company manufactures cable that sells for \$120 per meter. For a cable of fixed length, the cost of producing a portion of the cable varies with its distance from the beginning of the cable. Mighty reports that the cost to produce a portion of a cable that is  $x$  meters from the beginning of the cable is  $6\sqrt{x}$  dollars per meter. (Note: Profit is defined to be the difference between the amount of money received by the company for selling the cable and the company's cost of producing the cable.)

Find the maximum profit that Mighty could earn on the sale of one cable. Justify your answer.

## Part D

1 point is earned for  $P'(k)=0$

1 point is earned for  $k=400$

1 point is earned for answer

1 point is earned for justification

Let  $P(k)$  be the profit for a cable of length  $k$ .  $P'(k) = 120 - 6\sqrt{k} = 0$  when  $k=400$ .

This is the only critical point for  $P$ , and  $P'$  changes from positive to negative at  $k=400$ . Therefore, the maximum profit is  $P(400)=16,000$  dollars.



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns four of the following points:

1 point is earned for  $P'(k)=0$

1 point is earned for  $k=400$

1 point is earned for answer

1 point is earned for justification

Let  $P(k)$  be the profit for a cable of length  $k$ .

$P'(k) = 120 - 6\sqrt{k} = 0$  when  $k=400$ .

This is the only critical point for  $P$ , and  $P'$  changes from positive to negative at  $k=400$ .

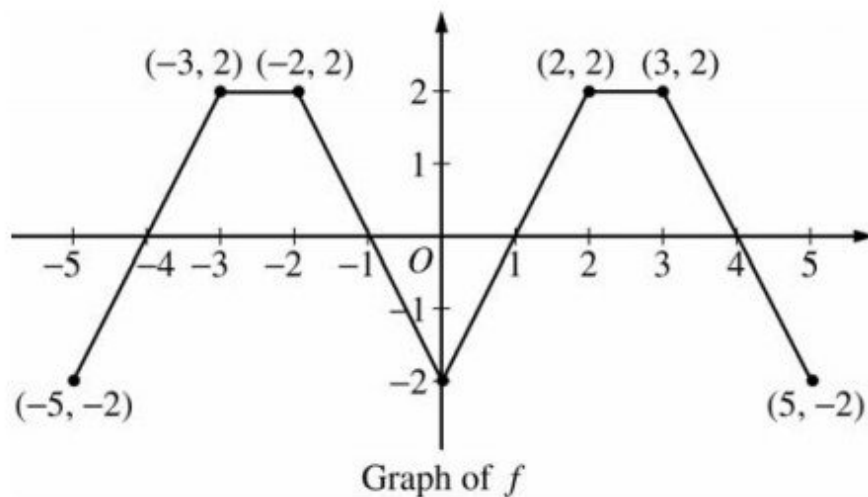
Therefore, the maximum profit is  $P(400)=16,000$  dollars.

5-4

69. If  $g$  is a differentiable function such that  $g(x) < 0$  for all real numbers  $x$  and if  $f'(x) = (x^2 - 4)g(x)$ , which of the following is true?
- (A)  $f$  has a relative maximum at  $x = -2$  and a relative minimum at  $x = 2$ .
  - (B)  $f$  has a relative minimum at  $x = -2$  and a relative maximum at  $x = 2$ . ✓
  - (C)  $f$  has relative minima at  $x = -2$  and at  $x = 2$ .
  - (D)  $f$  has relative maxima at  $x = -2$  and at  $x = 2$ .
  - (E) It cannot be determined if  $f$  has any relative extrema.

5-4

70.



The graph of the function  $f$  shown above consists of six line segments. Let  $g$  be the function given by  $g(x) = \int_0^x f(t) dt$ .

Does  $g$  have a relative minimum, a relative maximum, or neither at  $x = 1$ ? Justify your answer.

**Part B**

1 point is earned for the answer

1 point is earned for the reason

$g$  has a relative minimum at  $x = 1$

because  $g' = f$  changes from negative to positive at  $x = 1$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

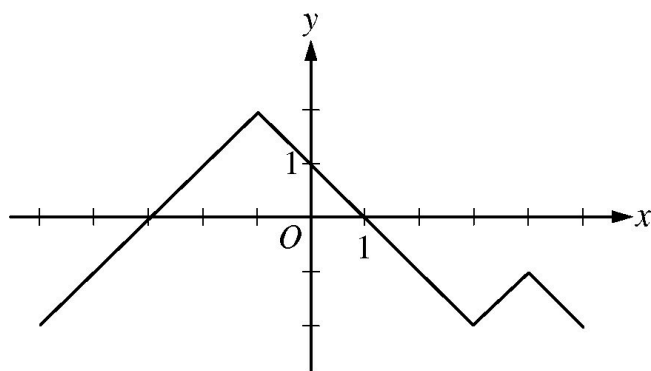
1 point is earned for the reason

$g$  has a relative minimum at  $x = 1$

because  $g' = f$  changes from negative to positive at  $x = 1$ .

5-4

71.

Graph of  $f'$ 

The graph of  $f'$ , the derivative of  $f$ , is shown in the figure above. The function  $f$  has a local maximum at  $x =$

(A) -3

(B) -1

(C) 1

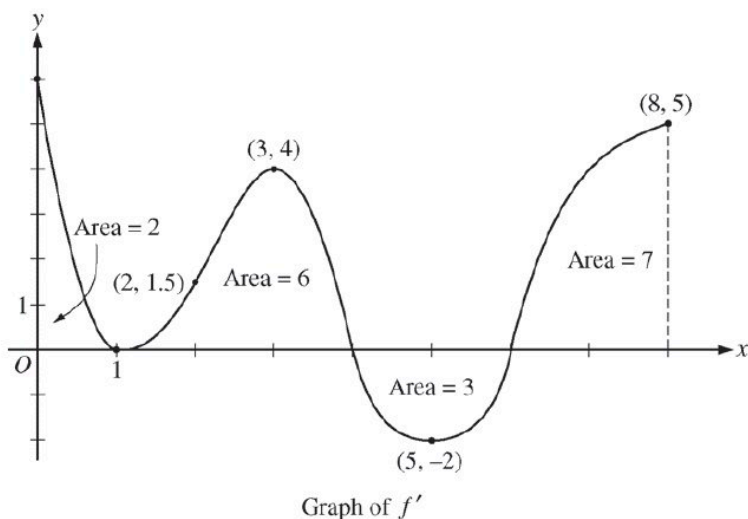
(D) 3

(E) 4



5-4

72.



The figure above shows the graph of  $f'$  the derivative of a twice-differentiable function  $f$ , on the closed interval  $0 \leq x \leq 8$ . The graph of  $f'$  has horizontal tangent lines at  $x=1$ ,  $x=3$ , and  $x=5$ . The areas of the regions between the graph of  $f'$  and the  $x$ -axis are labeled in the figure. The function  $f$  is defined for all real numbers and satisfied  $f(8)=4$ .

Find all values of  $x$  on the open interval  $0 < x < 8$  for which the function  $f$  has a local minimum. Justify your answer.

**Part A**

1 point is earned for the answer with justification

$x=6$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a local minimum at  $x=6$ .



|   |   |
|---|---|
| 0 | 1 |
|---|---|

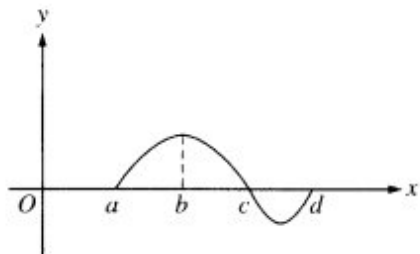
The student response earns all of the following points:

1 point is earned for the answer with justification

$x=6$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a local minimum at  $x=6$ .

5-4

73.



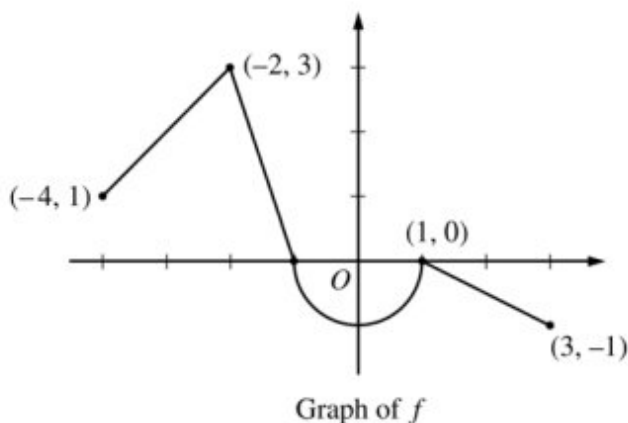
The graph of  $f$  is shown in the figure above. If  $g(x) = \int_a^x f(t)dt$ , for what value of  $x$  does  $g(x)$  have a maximum?

- (A)  $a$
- (B)  $b$
- (C)  $c$
- (D)  $d$
- (E) It cannot be determined from the information given.



5-4

74.



Let  $f$  be the continuous function defined on  $[-4, 3]$  whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let  $g$  be the function given by  $g(x) = \int_1^x f(t) dt$ .

Find the  $x$ -coordinate of each point at which the graph of  $g$  has a horizontal tangent line. For each Graph of  $f$  of these points, determine whether  $g$  has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

**Part C**

1 point is earned if considers  $g'(x)=0$

1 point is earned for  $x=-1$  and  $x=1$

1 point is earned for answers with justification

The graph of  $g$  has a horizontal tangent line where

$g'(x)=f(x)=0$ . This occurs at  $x=-1$  and  $x=1$ .  $g'(x)$  changes sign from positive to negative at  $x=-1$ . Therefore,  $g$  has a relative maximum at  $x=-1$ .  $g'(x)$  does not change sign at  $x=1$ . Therefore,  $g$  has neither a relative maximum nor a relative minimum at  $x=1$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned if considers  $g'(x)=0$

1 point is earned for  $x=-1$  and  $x=1$

1 point is earned for answers with justification

The graph of  $g$  has a horizontal tangent line where

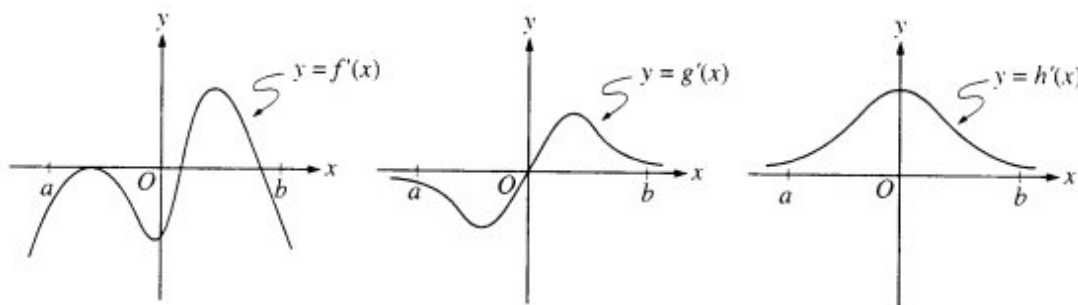
$g'(x)=f(x)=0$ . This occurs at  $x=-1$  and  $x=1$ .

5-4

$g'(x)$  changes sign from positive to negative at  $x=-1$ . Therefore,  $g$  has a relative maximum at  $x=-1$ .

$g'(x)$  does not change sign at  $x=1$ . Therefore,  $g$  has neither a relative maximum nor a relative minimum at  $x=1$ .

75.



The graphs of the derivatives of the functions  $f$ ,  $g$ , and  $h$  are shown above. Which of the functions  $f$ ,  $g$ , or  $h$  have a relative maximum on the open interval  $a < x < b$ ?

(A)  $f$  only



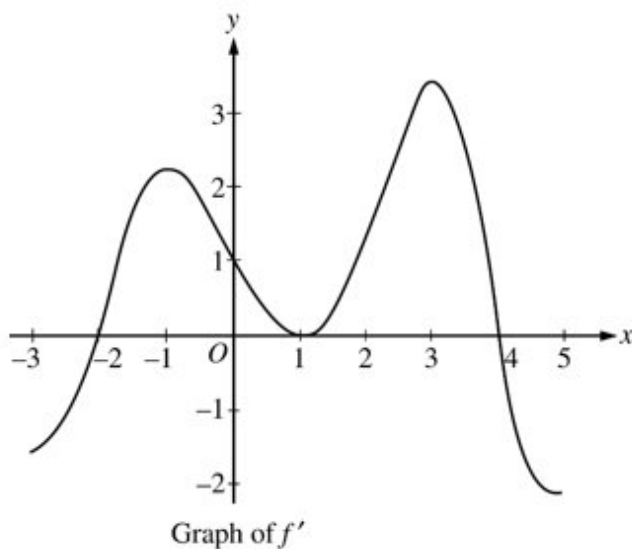
(B)  $g$  only

(C)  $h$  only

(D)  $f$  and  $g$  only

(E)  $f$ ,  $g$ , and  $h$

76.



The graph of the derivative of a function  $f$  is shown in the figure above. The graph has horizontal tangent lines at  $x = -1$ ,  $x = 1$ , and  $x = 3$ . At which of the following values of  $x$  does  $f$  have a relative maximum?

(A)  $-2$  only

(B)  $1$  only

(C)  $4$  only

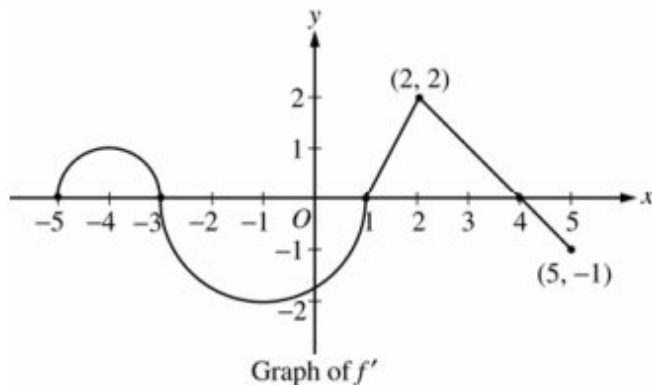


(D)  $-1$  and  $3$  only

(E)  $-2$ ,  $1$ , and  $4$

5-4

77.



Let  $f$  be a function defined on the closed interval  $-5 \leq x \leq 5$  with  $f(1)=3$ . The graph of  $f'$  the derivative of  $f$ , consists of two semicircles and two line segments, as shown above.

For  $-5 < x < 5$ , find all values  $x$  at which  $f$  has a relative maximum. Justify your answer.

**Part A**

1 point is earned for the  $x$ -values

1 point is earned for the justification

$f'(x)=0$  at  $x=-3, 1, 4$

$f'$  changes from positive to negative at  $-3$  and  $4$ .

Thus,  $f$  has a relative maximum at  $x=-3$  and at  $x=4$ .

✓

|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the  $x$ -values

1 point is earned for the justification

$f'(x)=0$  at  $x=-3, 1, 4$

$f'$  changes from positive to negative at  $-3$  and  $4$ .

Thus,  $f$  has a relative maximum at  $x=-3$  and at  $x=4$ .

5-4

78.

| $x$     | -2 | $-2 < x < -1$ | -1            | $-1 < x < 1$ | 1 | $1 < x < 3$ | 3             |
|---------|----|---------------|---------------|--------------|---|-------------|---------------|
| $f(x)$  | 12 | Positive      | 8             | Positive     | 2 | Positive    | 7             |
| $f'(x)$ | -5 | Negative      | 0             | Negative     | 0 | Positive    | $\frac{1}{2}$ |
| $g(x)$  | -1 | Negative      | 0             | Positive     | 3 | Positive    | 1             |
| $g'(x)$ | 2  | Positive      | $\frac{3}{2}$ | Positive     | 0 | Negative    | -2            |

The twice-differentiable functions  $f$  and  $g$  are defined for all real numbers  $x$ . Values of  $f$ ,  $f'$ ,  $g$ , and  $g'$  for various values of  $x$  are given in the table above.

Find the  $x$ -coordinate of each relative minimum of  $f$  on the interval  $[-2, 3]$ . Justify your answers.

**Part A**

1 point is earned for the answer with justification

$x=1$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a relative minimum at  $x=1$ .



|   |   |
|---|---|
| 0 | 1 |
|---|---|

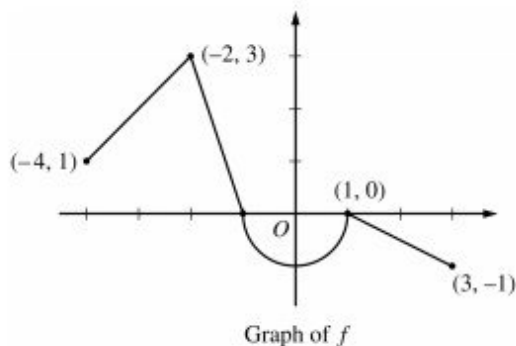
The student response earns one of the following points:

1 point is earned for the answer with justification

$x=1$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a relative minimum at  $x=1$ .

5-4

79.



Let  $f$  be the continuous function defined on  $[-4, 3]$  whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let  $g$  be the function given by  $g(x) = \int_1^x f(t) dt$ .

- Find the values of  $g(2)$  and  $g(-2)$ .
- For each of  $g'(-3)$  and  $g''(-3)$ , find the value or state that it does not exist.
- Find the  $x$ -coordinate of each point at which the graph of  $g$  has a horizontal tangent line. For each of these points, determine whether  $g$  has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For  $-4 < x < 3$ , find all values of  $x$  for which the graph of  $g$  has a point of inflection. Explain your reasoning.

**Part A**

1 point is earned for  $g(2)$

1 point is earned for  $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $g(2)$

1 point is earned for  $g(-2)$

5-4

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt$$

$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right)$$

**Part B**1 point is earned for  $g'(-3)$ 1 point is earned for  $g''(-3)$ 

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $g'(-3)$ 1 point is earned for  $g''(-3)$ 

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

**Part C**1 point is earned if considers  $g'(x) = 0$ 1 point is earned for  $x = -1$  and  $x = 1$ 

1 point is earned for answers with justification

The graph of  $g$  has a horizontal tangent line where  $g'(x) = f(x) = 0$ . This occurs at  $x = -1$  and  $x = 1$ . $g'(x)$  changes sign from positive to negative at  $x = -1$ . Therefore,  $g$  has a relative maximum at  $x = -1$ . $g'(x)$  does not change sign at  $x = 1$ . Therefore,  $g$  has neither a relative maximum nor a relative minimum at  $x = 1$ .

5-4



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned if considers  $g'(x) = 0$

1 point is earned for  $x = -1$  and  $x = 1$

1 point is earned for answers with justification

The graph of  $g$  has a horizontal tangent line where  $g'(x) = f(x) = 0$ . This occurs at  $x = -1$  and  $x = 1$ .

$g'(x)$  changes sign from positive to negative at  $x = -1$ . Therefore,  $g$  has a relative maximum at  $x = -1$ .

$g'(x)$  does not change sign at  $x = 1$ . Therefore,  $g$  has neither a relative maximum nor a relative minimum at  $x = 1$ .

#### Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of  $g$  has a point of inflection at each of  $x = -2$ ,  $x = 0$ , and  $x = 1$  because  $g''(x) = f'(x)$  changes sign at each of these values.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of  $g$  has a point of inflection at each of  $x = -2$ ,  $x = 0$ , and  $x = 1$  because  $g''(x) = f'(x)$  changes sign at each of these values.

## 5-4

80. A particle moves along the  $x$ -axis so that at any time  $t > 0$  its velocity is given by  $v(t) = t \ln t - t$ . At time  $t = 1$ , the position of the particle is  $x(1) = 6$ .
- Write an expression for the acceleration of the particle.
  - For what values of  $t$  is the particle moving to the right?
  - What is the minimum velocity of the particle? Show the analysis that leads to your conclusion.
  - Write an expression of the position  $x(t)$  of the particle.

## Part A

1 point is earned for the correct  $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student earns all of the following points:

1 point is earned for the correct  $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

## Part B

1 point is earned for correctly setting  $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

$$= t(\ln t - 1) > 0$$

1 point is earned for the correct answer

$$t > e$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for correctly setting  $t \ln t - t > 0$

$$v(t) = t \ln t - t > 0$$

$$= t(\ln t - 1) > 0$$

**5-4**

1 point is earned for the correct answer

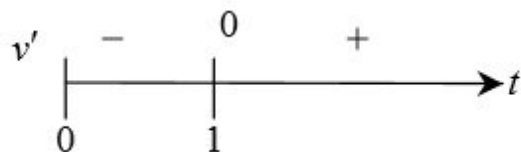
$$t > e$$

**Part C**1 point is earned for correctly solving  $v'(t) = 0$ 

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is  $v(1) = -1$ Max 1/3 if student's  $v'$  does not involve  $\ln$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

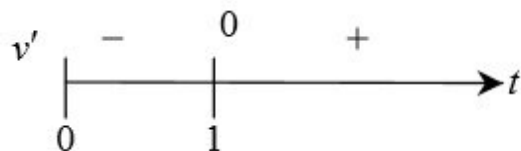
The student response earns all of the following points:

1 point is earned for correctly solving  $v'(t) = 0$ 

$$v'(t) = \ln t = 0$$

$$t = 1$$

1 point is earned for the correct analysis



1 point is earned for the correct answer

minimum velocity is  $v(1) = -1$

## 5-4

Max 1/3 if student's  $v'$  does not involve  $\ln$

**Part D**

2 points are earned for the correct antiderivative of  $t \ln t - t$

<-2>: error in  $\int$  -by-parts

<-1>: any other error

$$\begin{aligned} \int t \ln t - t \, dt &= \left( \frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 \right) - \frac{1}{2} t^2 + C \\ &= \frac{1}{2} t^2 \ln t - \frac{3}{4} t^2 + C; \\ 6 = x(1) &= 0 - \frac{3}{4} + C \end{aligned}$$

1 point is earned for correctly solving for  $C$

$$\begin{aligned} C &= \frac{27}{4}; \\ x(t) &= \frac{1}{2} t^2 \ln t - \frac{3}{4} t^2 + \frac{27}{4} \end{aligned}$$

0/3 if not antidifferentiating  $t \ln t - t$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for the correct antiderivative of  $t \ln t - t$

<-2>: error in  $\int$  -by-parts

<-1>: any other error

$$\begin{aligned} \int t \ln t - t \, dt &= \left( \frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 \right) - \frac{1}{2} t^2 + C \\ &= \frac{1}{2} t^2 \ln t - \frac{3}{4} t^2 + C; \\ 6 = x(1) &= 0 - \frac{3}{4} + C \end{aligned}$$

**5-4**

1 point is earned for correctly solving for  $C$

$$C = \frac{27}{4};$$

$$x(t) = \frac{1}{2}t^2 \ln t - \frac{3}{4}t^2 + \frac{27}{4}$$

0/3 if not antidifferentiating  $t \ln t - t$

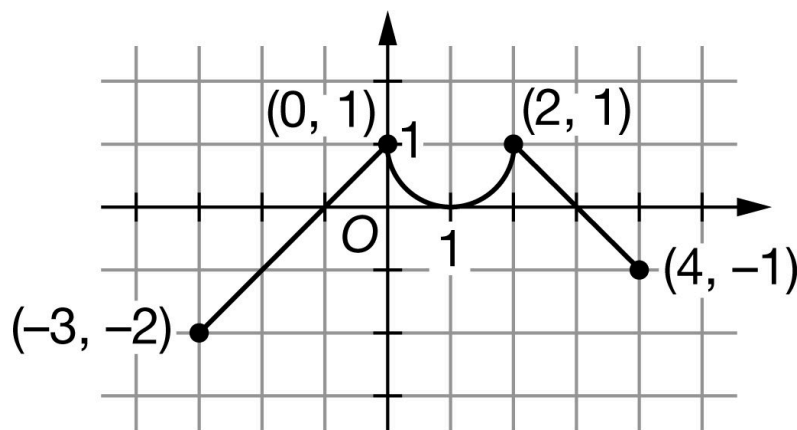
5-4

## 81. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function  $f$  is assumed to be the set of all real numbers  $x$  for which  $f(x)$  is a real number.

Graph of  $g$ 

The continuous function  $g$  has domain  $-3 \leq x \leq 4$ . The graph of  $g$ , consisting of two line segments and a semicircle, is shown in the figure above. The graph of  $g$  has a horizontal tangent at  $x = 1$ . Let  $h$  be the

function defined by  $h(x) = \int_0^x g(t) \, dt$  for  $-3 \leq x \leq 4$ .

- Find the  $x$ -coordinate of each critical point of  $h$  on the interval  $-3 < x < 4$ .
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for  $h$ . Justify each of your answers.
- For  $-3 < x < 4$ , on what open intervals is  $h$  decreasing and concave down? Give a reason for your answer.
- Find the value of  $h(2)$ . Show the computations that lead to your answer.
- Find the value of  $h(-3)$ . Show the computations that lead to your answer.
- Find the value of  $h''(-1)$ , or explain why it does not exist.
- Must there exist a value of  $r$ , for  $-3 < r < 1$ , such that  $h'(r)$  is equal to the average rate of change of  $h$  over the interval  $-3 \leq x \leq 1$ ? Justify your answer.
- Find  $\lim_{x \rightarrow 2} \frac{h'(x)}{5x}$ . Show the computations that lead to your answer.
- The function  $k$  is defined by  $k(x) = h(x^2)$ . Find  $k'(-2)$ . Show the computations that lead to your answer.

## Part A

## 5-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values:  $x = -1$ ,  $x = 1$ , and  $x = 3$ .

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval  $(-3, 4)$ , or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval  $(-3, 4)$ . The point is still earned provided the three critical points  $x = -1$ ,  $x = 1$ , and  $x = 3$  are correctly listed.

If the critical points are given as ordered pairs, e.g.,  $(-1, 0)$ ,  $(1, 0)$ , and  $(3, 0)$ , read the  $x$ -coordinates, and disregard the  $y$ -coordinates.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes the criteria below.

☐ Answer

**Solution:**

$h'(x) = g(x) = 0$  at  $x = -1$ ,  $x = 1$ , and  $x = 3$ .

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on  $h'$  is eligible for all three points.

If a response gives justifications based only on  $g$ , then:

- the response is eligible for all three points, if the response presents the connection between  $h'(x)$  and  $g(x)$ , (that is,  $h' = g$ ). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of  $g$  or  $h'$  as  $h' = g$ , throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection  $h'(x) = g(x)$ . Any additional error (for example, not justifying that at  $x = -1$ ,  $h$  has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to  $g(x)$  as indicated by the stem.

## 5-4

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with  $h'$  produces a response that would earn all three points.

A response that discusses  $h'$  at a particular point (e.g., “ $h'(3)$  changes from negative to positive”) when the discussion should have been about  $h'(x)$  can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points.  $h(x)$  values for  $x = -1, 1, 3$  must be included in the table or list of values. The correct values of relevant inputs include:

- $h(-3) = \frac{3}{2}$
- $h(-1) = -\frac{1}{2}$
- $h(1) = 1 - \frac{\pi}{4}$
- $h(3) = \frac{5}{2} - \frac{\pi}{2}$
- $h(4) = 2 - \frac{\pi}{2}$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At  $x = -1$ ,  $h$  has a relative minimum because  $h' = g$  is negative for  $x < -1$  and  $h' = g$  is positive for  $x > -1$ .
- At  $x = 1$ ,  $h$  has neither because  $h' = g$  is positive for  $x < 1$  and  $h' = g$  is positive for  $x > 1$ .
- At  $x = 3$ ,  $h$  has a relative maximum because  $h' = g$  is positive for  $x < 3$  and  $h' = g$  is negative for  $x > 3$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

## 5-4

**Solution:**

At  $x = -1$ ,  $h$  has a relative minimum because  $h'(x) = g(x)$  changes from negative to positive there.

At  $x = 1$ ,  $h$  has neither because  $h'(x) = g(x)$  does not change sign there.

At  $x = 3$ ,  $h$  has a relative maximum because  $h'(x) = g(x)$  changes from positive to negative there.

**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is,  $[3, 4]$ ,  $(3, 4]$ ,  $[3, 4)$ , and  $(3, 4)$  all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only  $h'$  is eligible to earn the second point.

A response referencing only  $g$  is not eligible to earn the second point, unless the response has established the link between  $g(x)$  and  $h'(x)$  previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ $h'$  is decreasing” can also be expressed by either “ $h''$  is negative” or “the slope of  $h'$  is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

**Solution:**

$h$  is decreasing where  $h' = g$  is negative.

$h$  is concave down where  $h' = g$  is decreasing.

$h$  is decreasing and concave down on the open interval  $(3, 4)$ .

**Part D**

## 5-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is  $h(2) = 0.429$ . If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes the criteria below.

☐ Answer

**Solution:**

$$h(2) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of  $-\frac{1}{2} + 2$  earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is,  $-2$  or  $\frac{1}{2}$ ) does not need to be explicitly linked with an integral to earn the first point.

A response of  $h(-3) = \frac{3}{2}$  with no supporting work earns the second point only.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes **both** of the criteria below.

☐  $\int_{-3}^{-1} g(t) dt = -2$  – OR –  $\int_{-1}^0 g(t) dt = \frac{1}{2}$

☐ Answer

**Solution:**

## 5-4

$$\begin{aligned}
 h(-3) &= \int_0^{-3} g(t) \, dt = - \int_{-3}^0 g(t) \, dt \\
 &= - \int_{-3}^{-1} g(t) \, dt - \int_{-1}^0 g(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

**Part F**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of  $h''(-1) = 1$  or  $g'(-1) = 1$  or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes the criteria below.

☐ Answer

**Solution:**

$$h''(x) = g'(x)$$

$$h''(-1) = g'(-1) = \frac{1 - (-2)}{0 - (-3)} = 1$$

**Part G**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that  $h$  is continuous and differentiable. An ambiguous reference to  $h$  (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which  $h$  is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes,  $h(x)$  is continuous and differentiable so MVT applies” earns both points.
- “Yes,  $h(x)$  is continuous and differentiable, therefore there must be a value where  $h'(r)$  equals the average rate of change of  $h'(x)$ ” earns both points.

## 5-4

- “Yes,  $h'(x)$  (the derivative of  $h(x)$ ) is continuous and differentiable, therefore there must be a value where  $h'(r)$  equals the average rate of change of  $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because  $h(x)$  is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because  $h(x)$  is continuous” earns neither point.
- “Yes, because  $h(x)$  is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ $h$  is not differentiable, so therefore the MVT does not apply” earns one of the two points.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

**Solution:**

$h' = g \Rightarrow h$  is differentiable on  $-3 \leq x \leq 1 \Rightarrow h$  is continuous on  $-3 \leq x \leq 1$ .

Therefore, the Mean Value Theorem can be applied to  $h$  on the interval  $-3 \leq x \leq 1$  to guarantee that there exists a value of  $r$ , for  $-3 < r < 1$ , such that  $h'(r)$  equals the average rate of change of  $h$  over the interval  $-3 \leq x \leq 1$ .

**Part H**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of  $\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{1}{10}$  earns the point, as does just stating the answer of  $\frac{1}{10}$ .

No additional supporting work is required.

The response does not need to state that  $h'$  is continuous. If the response does state that  $h'$  is continuous, an explanation is not required.

5-4



0

1

The student response accurately includes the criteria below.

☐ Answer

**Solution:**

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} h'(x) = h'(2)$$

$$\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{h'(2)}{5 \cdot 2} = \frac{g(2)}{10} = \frac{1}{10}$$

**Part I**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If  $x = -2$  is substituted for  $x$ , then  $x = -2$  must appear unsimplified in at least one factor in order to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot h'(4) \cdot 2 \cdot (-2)$$

$$\cdot g(4) \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4)$$

$$\cdot g((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

## 5-4

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(4) \cdot (-4)$$

$$\cdot g(4) \cdot (-4)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking  $h'$  and  $g$  either directly or indirectly in the presented derivative of  $k(x)$  or in  $k'(-2)$ . For example, the second point is earned by  $g(x^2)$  or by  $g((-2)^2)$ .

The second point may be earned without earning the first point. For example,  $h'(x^2) \cdot 2 = g(x^2) \cdot 2$  does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because  $h'$  and  $Ag$  are linked.

The second point may be earned by going directly from  $h'(4)$  to  $-1$ . The function  $g$  does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2) = -1 \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4) = -1 \cdot (-4)$$

Recall, if  $x = -2$  is substituted for  $x$ , then  $x = -2$  must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both  $g(x^2) \cdot 2x$  and  $g((-2)^2) \cdot 2 \cdot (-2)$  earn the first two points.

The third point is earned for the correct answer of 4 simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form  $h'(x)^2 \cdot 2x$  or  $(h'(x))^2 \cdot 2x$  are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

**5-4****Solution:**

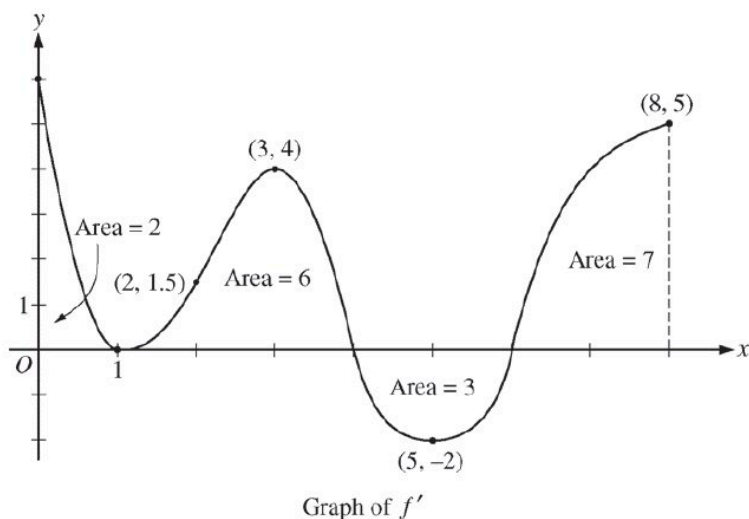
$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-2) = g((-2)^2) \cdot 2 \cdot (-2) = g(4) \cdot (-4) = -1 \cdot (-4) = 4$$

5-4

82.



The figure above shows the graph of  $f'$ , the derivative of a twice-differentiable function  $f$ , on the closed interval  $0 \leq x \leq 8$ . The graph of  $f'$  has horizontal tangent lines at  $x = 1$ ,  $x = 3$ , and  $x = 5$ . The areas of the regions between the graph of  $f'$  and the  $x$ -axis are labeled in the figure. The function  $f$  is defined for all real numbers and satisfies  $f(8) = 4$ .

- Find all values of  $x$  on the open interval  $0 < x < 8$  for which the function  $f$  has a local minimum. Justify your answer.
- Determine the absolute minimum value of  $f$  on the closed interval  $0 \leq x \leq 8$ . Justify your answer.
- On what open intervals contained in  $0 < x < 8$  is the graph of  $f$  both concave down and increasing? Explain your reasoning.
- The function  $g$  is defined by  $g(x) = (f(x))^3$ . If  $f(3) = -5/2$ , find the slope of the line tangent to the graph of  $g$  at  $x = 3$ .

### Part A

1 point is earned for the answer with justification

$x = 6$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a local minimum at  $x = 6$ .



0

1

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 6$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a local minimum at  $x = 6$ .

### Part B

## 5-4

1 point is earned for considering  $x = 0$  and  $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at  $x = 6$  or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \end{aligned}$$

$$\begin{aligned} f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

$$f(8) = 4$$

The absolute minimum value of  $f$  on the closed interval  $[0, 8]$  is  $-8$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for considering  $x = 0$  and  $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at  $x = 6$  or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \end{aligned}$$

$$\begin{aligned} f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

**5-4**

$$f(8) = 4$$

The absolute minimum value of  $f$  on the closed interval  $[0, 8]$  is -8.

**Part C**

1 point is earned for the answer

1 point is earned for the explanation

The graph of  $f$  is concave down and increasing on  $0 < x < 1$  and  $3 < x < 4$ , because  $f'$  is decreasing and positive on these intervals.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of  $f$  is concave down and increasing on  $0 < x < 1$  and  $3 < x < 4$ , because  $f'$  is decreasing and positive on these intervals.

**Part D**

2 points are earned for  $g'(x)$

1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for  $g'(x)$

1 point is earned for the answer

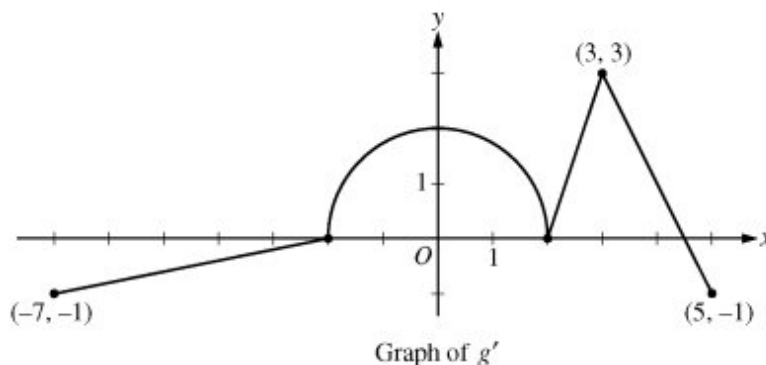
**5-4**

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$

5-4

83.



The function  $g$  is defined and differentiable on the closed interval  $[-7, 5]$  and satisfies  $g(0) = 5$ . The graph of  $y = g'(x)$ , the derivative of  $g$ , consists of a semicircle and three line segments, as shown in the figure above.

- (a) Find  $g(3)$  and  $g(-2)$ .
- (b) Find the  $x$ -coordinate of each point of inflection of the graph of  $y = g(x)$  on the interval  $-7 < x < 5$ . Explain your reasoning.
- (c) The function  $h$  is defined by  $h(x) = g(x) - \frac{1}{2}x^2$ . Find the  $x$ -coordinate of each critical point of  $h$ , where  $-7 < x < 5$ , and classify each critical point as the location of a relative minimum, relative maximum, or neither a minimum nor a maximum. Explain your reasoning.

**Part A**

1 point is earned for using  $g(0) = 5$

1 point is earned for  $g(3)$

1 point is earned for  $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for using  $g(0) = 5$

1 point is earned for  $g(3)$

1 point is earned for  $g(-2)$

5-4

$$g(3) = 5 + \int_0^3 g'(x)dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x)dx = 5 - \pi$$

**Part B**

1 point is earned for identifying  $x = 0, 2, 3$

1 point is earned for explanation

The graph of  $y = g(x)$  has points of inflection at  $x = 0, x = 2$ , and  $x = 3$  because  $g'$  changes from increasing to decreasing at  $x = 0$  and  $x = 3$ , and  $g'$  changes from decreasing to increasing at  $x = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for identifying  $x = 0, 2, 3$

1 point is earned for explanation

The graph of  $y = g(x)$  has points of inflection at  $x = 0, x = 2$ , and  $x = 3$  because  $g'$  changes from increasing to decreasing at  $x = 0$  and  $x = 3$ , and  $g'$  changes from decreasing to increasing at  $x = 2$ .

**Part C**

1 point is earned for  $h'(x)$

1 point is earned for identifying  $x = \sqrt{2}, 3$

1 point is earned for answer for  $\sqrt{2}$  with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval  $-2 \leq x \leq 2$ ,  $g'(x) = \sqrt{4 - x^2}$ .

5-4

On this interval,  $g'(x) = x$  when  $x = \sqrt{2}$ .

The only other solution to  $g'(x) = x$  is  $x = 3$ .

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore  $h$  has a relative maximum at  $x = \sqrt{2}$ , and  $h$  has neither a minimum nor a maximum at  $x = 3$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $h'(x)$

1 point is earned for identifying  $x = \sqrt{2}$ , 3

1 point is earned for answer for  $\sqrt{2}$  with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval  $-2 \leq x \leq 2$ ,  $g'(x) = \sqrt{4 - x^2}$ .

On this interval,  $g'(x) = x$  when  $x = \sqrt{2}$ .

The only other solution to  $g'(x) = x$  is  $x = 3$ .

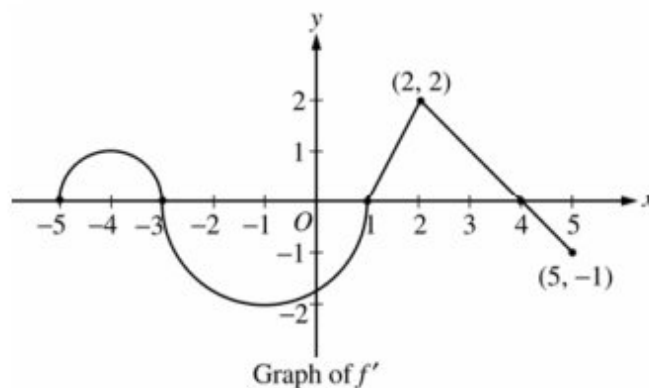
$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore  $h$  has a relative maximum at  $x = \sqrt{2}$ , and  $h$  has neither a minimum nor a maximum at  $x = 3$ .

5-4

84.



Let  $f$  be a function defined on the closed interval  $-5 \leq x \leq 5$  with  $f(1) = 3$ . The graph of  $f'$  the derivative of  $f$ , consists of two semicircles and two line segments, as shown above.

- For  $-5 < x < 5$ , find all values  $x$  at which  $f$  has a relative maximum. Justify your answer.
- For  $-5 < x < 5$ , find all values  $x$  at which the graph of  $f$  has a point of inflection. Justify your answer.
- Find all intervals on which the graph of  $f$  is concave up and also has positive slope. Explain your reasoning.
- Find the absolute minimum value of  $f(x)$  over the closed interval  $-5 \leq x \leq 5$ . Explain your reasoning.

**Part A**

1 point is earned for the  $x$ -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

$f'$  changes from positive to negative at -3 and 4.

Thus,  $f$  has a relative maximum at  $x = -3$  and at  $x = 4$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the  $x$ -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

$f'$  changes from positive to negative at -3 and 4.

## 5-4

Thus,  $f$  has a relative maximum at  $x = -3$  and at  $x = 4$ .

**Part B**

1 point is earned for the  $x$ -values

1 point is earned for the justification

$f'$  changes from increasing to decreasing, or vice versa, at  $x = -4$ ,  $-1$ , and  $2$ . Thus, the graph of  $f$  has points of inflection when  $x = -4$ ,  $-1$ , and  $2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the  $x$ -values

1 point is earned for the justification

$f'$  changes from increasing to decreasing, or vice versa, at  $x = -4$ ,  $-1$ , and  $2$ . Thus, the graph of  $f$  has points of inflection when  $x = -4$ ,  $-1$ , and  $2$ .

**Part C**

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of  $f$  is concave up with positive slope where  $f'$  is increasing and positive:  $-5 < x < -4$  and  $1 < x < 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of  $f$  is concave up with positive slope where  $f'$  is increasing and positive:  $-5 < x < -4$  and  $1 < x < 2$ .

**Part D**

1 point is earned if identifies  $x = 1$  as a candidate

## 5-4

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where  $f'$  changes from negative to positive (at  $x = 1$ ) and at the endpoints ( $x = -5, 5$ ).

$$f(-5) = 3 + \int_1^{-5} f(x) \, dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x) \, dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of  $f$  on  $[-5, 5]$  is  $f(1) = 3$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned if identifies  $x = 1$  as a candidate

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where  $f'$  changes from negative to positive (at  $x = 1$ ) and at the endpoints ( $x = -5, 5$ ).

$$f(-5) = 3 + \int_1^{-5} f(x) \, dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

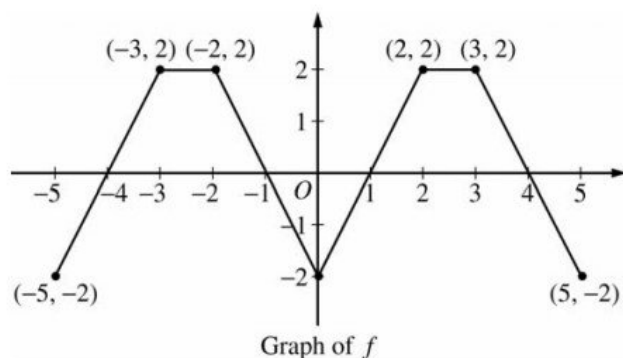
$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x) \, dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of  $f$  on  $[-5, 5]$  is  $f(1) = 3$ .

5-4

85.



The graph of the function  $f$  shown above consists of six line segments. Let  $g$  be the function given by

$$g(x) = \int_0^x f(t) dt.$$

(a) Find  $g(4)$ ,  $g'(4)$ , and  $g''(4)$ .

(b) Does  $g$  have a relative minimum, a relative maximum, or neither at  $x = 1$ ? Justify your answer.

(c) Suppose that  $f$  is defined for all real numbers  $x$  and is periodic with a period of length 5. The graph above shows two periods of  $f$ . Given that  $g(5) = 2$ , find  $g(10)$  and write an equation for the line tangent to the graph of  $g$  at  $x = 108$ .

### Part A

1 point is earned for:  $g(4)$

1 point is earned for:  $g'(4)$

1 point is earned for:  $g''(4)$

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $g(4)$

1 point is earned for:  $g'(4)$

1 point is earned for:  $g''(4)$

5-4

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$

**Part B**

1 point is earned for the answer

1 point is earned for the reason

$g$  has a relative minimum at  $x = 1$

because  $g' = f$  changes from negative to positive at  $x = 1$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

$g$  has a relative minimum at  $x = 1$

because  $g' = f$  changes from negative to positive at  $x = 1$ .

**Part C**

1 point is earned for:  $g(10)$

1 point is earned for:  $g(108)$

1 point is earned for:  $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$  and the function values of  $g$  increase by 2 for every increase of 5 in  $x$ .

$$g(10) = 2g(5) = 4$$

5-4

$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of  $g$  at  $x = 108$  is  $y - 44 = 2(x - 108)$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $g(10)$

1 point is earned for:  $g(108)$

1 point is earned for:  $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$  and the function values of  $g$  increase by 2 for every increase of 5 in  $x$ .

$$g(10) = 2g(5) = 4$$

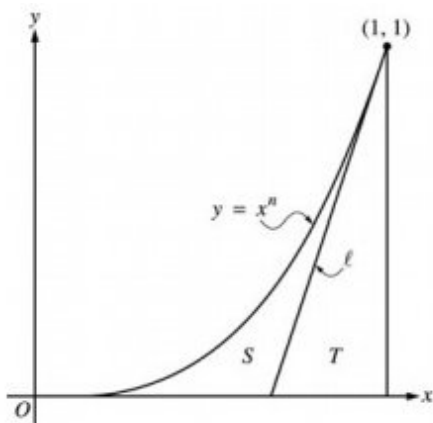
$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of  $g$  at  $x = 108$  is  $y - 44 = 2(x - 108)$ .

5-4

86.



Let  $l$  be the line tangent to the graph of  $y = x^n$  at the point  $(1, 1)$ , where  $n > 1$ , as shown above.

(a) Find  $\int_0^1 x^n dx$  in terms of  $n$ .

(b) Let  $T$  be the triangular region bounded by  $l$ , the  $x$ -axis, and the line  $x = 1$ . Show that the area of  $T$  is  $1/2n$ .

(c) Let  $S$  be the region bounded by the graph of  $y = x^n$ , the line  $l$ , and the  $x$ -axis. Express the area of  $S$  in terms of  $n$  and determine the value of  $n$  that maximizes the area of  $S$ .

**Part A**

1 point is earned for the antiderivative of  $x^n$

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for the antiderivative of  $x^n$

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$

5-4

**Part B**

1 point is earned if slope of line  $l$  is  $n$

1 point is earned if base of  $T$  is  $\frac{1}{n}$

1 point is earned if shows area is  $\frac{1}{2n}$

Let  $b$  be the length of the base of triangle  $T$ .

$\frac{1}{b}$  is the slope of line  $l$ , which is  $n$

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned if slope of line  $l$  is  $n$

1 point is earned if base of  $T$  is  $\frac{1}{n}$

1 point is earned if shows area is  $\frac{1}{2n}$

Let  $b$  be the length of the base of triangle  $T$ .

$\frac{1}{b}$  is the slope of line  $l$ , which is  $n$

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$

**Part C**

## 5-4

1 point is earned for the area of  $S$  in terms of  $n$

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for  $n$

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for the area of  $S$  in terms of  $n$

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for  $n$

$$\begin{aligned}\text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\ &= \frac{1}{n+1} - \frac{1}{2n} \\ \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\ 2n^2 &= (n+1)^2 \\ \sqrt{2}n &= (n+1) \\ n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}\end{aligned}$$

5-4

87.  A particle moves along the  $y$ -axis so that its velocity  $v$  at time  $t \geq 0$  is given by  $v(t) = 1 - \tan^{-1}(e^t)$ . At time  $t = 0$ , the particle is at  $y = -1$ . (Note:  $\tan^{-1}x = \arctan x$ )
- Find the acceleration of the particle at time  $t = 2$ .
  - Is the speed of the particle increasing or decreasing at time  $t = 2$ ? Give a reason for your answer.
  - Find the time  $t \geq 0$  at which the particle reaches its highest point. Justify your answer.
  - Find the position of the particle at time  $t = 2$ . Is the particle moving toward the origin or away from the origin at time  $t = 2$ ? Justify your answer.

**Part A**

1 point is earned for the answer

$$a(2) = v'(2) = -0.132 \text{ or } -0.133$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the answer

$$a(2) = v'(2) = -0.132 \text{ or } -0.133$$

**Part B**

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since  $a(2) < 0$  and  $v(2) < 0$ .

|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the answer with reason

$$v(2) = -0.436$$

Speed is increasing since  $a(2) < 0$  and  $v(2) < 0$ .**Part C**

## 5-4

1 point is earned for: set  $v(t) = 0$

1 point is earned for identifying  $t = 0.433$  as a candidate

1 point is earned for justifying absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$  is the only critical value for  $y$ .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$  has an absolute maximum at  $t = 0.433$ .



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for: set  $v(t) = 0$

1 point is earned for identifying  $t = 0.433$  as a candidate

1 point is earned for justifying absolute maximum

$$v(t) = 0 \text{ when } \tan^{-1}(e^t) = 1$$

$t = \ln(\tan(1)) = 0.433$  is the only critical value for  $y$ .

$$v(t) > 0 \text{ for } 0 < t < \ln(\tan(1))$$

$$v(t) < 0 \text{ for } t > \ln(\tan(1))$$

$y(t)$  has an absolute maximum at  $t = 0.433$ .

**Part D**

1 point is earned for:  $\int_0^2 v(t) dt$

1 point is earned for the handles initial condition

1 point is earned for the value of  $y(2)$

1 point is earned for the answer with reason

5-4

$$y(2) = -1 + \int_0^2 v(t) dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since  $v(2) < 0$  and  $y(2) < 0$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for:  $\int_0^2 v(t) dt$

1 point is earned for the handles initial condition

1 point is earned for the value of  $y(2)$

1 point is earned for the answer with reason

$$y(2) = -1 + \int_0^2 v(t) dt = -1.360 \text{ or } -1.361$$

The particle is moving away from the origin since  $v(2) < 0$  and  $y(2) < 0$ .

5-4

88.

| $x$     | -2 | $-2 < x < -1$ | -1            | $-1 < x < 1$ | 1 | $1 < x < 3$ | 3             |
|---------|----|---------------|---------------|--------------|---|-------------|---------------|
| $f(x)$  | 12 | Positive      | 8             | Positive     | 2 | Positive    | 7             |
| $f'(x)$ | -5 | Negative      | 0             | Negative     | 0 | Positive    | $\frac{1}{2}$ |
| $g(x)$  | -1 | Negative      | 0             | Positive     | 3 | Positive    | 1             |
| $g'(x)$ | 2  | Positive      | $\frac{3}{2}$ | Positive     | 0 | Negative    | -2            |

The twice-differentiable functions  $f$  and  $g$  are defined for all real numbers  $x$ . Values of  $f$ ,  $f'$ ,  $g$ , and  $g'$  for various values of  $x$  are given in the table above.

- (a) Find the  $x$ -coordinate of each relative minimum of  $f$  on the interval  $[-2, 3]$ . Justify your answers.  
 (b) Explain why there must be a value  $c$ , for  $-1 < c < 1$ , such that  $f''(c) = 0$ .  
 (c) The function  $h$  is defined by  $h(x) = \ln(f(x))$ . Find  $h'(3)$ . Show the computations that lead to your answer.

(d) Evaluate  $\int_{-2}^3 f'(g(x))g'(x) dx$ .

**Part A**

1 point is earned for the answer with justification

$x = 1$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a relative minimum at  $x = 1$ .



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 1$  is the only critical point at which  $f'$  changes sign from negative to positive. Therefore,  $f$  has a relative minimum at  $x = 1$ .

**Part B**

1 point is earned for  $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

$f$  is differentiable  $\Rightarrow f'$  is continuous on the interval  $-1 \leq x \leq 1$

$$\frac{f'(1) - f'(-1)}{1 - (-1)} = \frac{0 - 0}{2} = 0$$

## 5-4

Therefore, by the Mean Value Theorem, there is at least one value  $c$ ,  $-1 < c < 1$ , such that  $f''(c) = 0$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

$f'$  is differentiable  $\Rightarrow f'$  is continuous on the interval  $-1 \leq x \leq 1$

$$\frac{f'(1) - f'(-1)}{1 - (-1)} = \frac{0 - 0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value  $c$ ,  $-1 < c < 1$ , such that  $f''(c) = 0$ .

**Part C**

2 points are earned for  $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for  $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

**Part D**

5-4

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}
 \int_{-2}^3 f'(g(x))g'(x) \, dx &= [f(g(x))]_{x=-2}^{x=3} \\
 &= f(g(3)) - f(g(-2)) \\
 &= f(1) - f(-1) \\
 &= 2 - 8 = -6
 \end{aligned}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}
 \int_{-2}^3 f'(g(x))g'(x) \, dx &= [f(g(x))]_{x=-2}^{x=3} \\
 &= f(g(3)) - f(g(-2)) \\
 &= f(1) - f(-1) \\
 &= 2 - 8 = -6
 \end{aligned}$$